

正修科技大學
工業工程與管理碩士班
碩士論文

ESG 風險管理與企業財務績效：新興市場中企業規模
之調節效果

**ESG Risk Management and Corporate Financial
Performance: The Contingent Role of Firm Size in an
Emerging Market**

研究 生：ARTI FUDIAH 富得運

指導教授：Cheng-Kang Lee 李政鋼

中華民國一百一十五年五月

CHENG SHIU UNIVERSITY

Department of Industrial Engineering and Management

Master Thesis

**ESG Risk Management and Corporate Financial Performance: The
Contingent Role of Firm Size in an Emerging Market**

The logo of Cheng Shiu University is a circular emblem. The outer ring is pink at the top and blue at the bottom, containing the text "CHENG SHIU UNIVERSITY" in white. The inner circle is white and features a stylized blue flame or leaf shape. The Chinese characters "正修科技大學" are written in white along the bottom arc of the inner circle.

Graduate Student: ARTI FUDIAH
Advisor : Prof. Cheng-Kang Lee

2026/5

正修科技大學

碩士學位考試委員會審定書

ESG 風險管理與企業財務績效：新興市場中企業規模之調節效果

ESG Risk Management and Corporate Financial Performance:

The Contingent Role of Firm Size in an Emerging Market

本論文係富得運君（M1206404）於本校 工業工程與管理系完成之碩士學位論文，於
民國 115 年 05 月 09 日承下列學位考試委員審查通過及口試及格，特此證明。

考試委員：

鄭永長

李政鋼

鄭若即

召集人：

鄭永長

指導教授：

李政鋼

系(所)主任：

葉俊賢

中華民國 115 年 05 月 09 日

摘要

本研究旨在分析環境、社會與公司治理（Environmental, Social, and Governance, ESG）績效、智慧資本、槓桿程度與流動性對企業績效之影響，並進一步探討公司規模在其中所扮演之調節角色。企業績效分別以總資產報酬率（Return on Assets, ROA）作為短期績效指標，以及托賓 Q（Tobin's Q）作為長期績效指標加以衡量。本研究採用次級資料，資料來源為 2021 年至 2023 年期間列入印尼證券交易所 ESG Leaders 指數（IDX ESG Leaders, IDXESGL）之上市公司財務報表。研究樣本係透過立意抽樣法選取，最終取得 58 家公司，共計 174 筆縱橫資料觀測值。

本研究採用縱橫資料迴歸分析，並以固定效果模型（Fixed Effects Model, FEM）進行估計；同時運用調節迴歸分析（Moderated Regression Analysis, MRA）檢驗公司規模之調節效果。ESG 績效係以轉換後之 ESG 風險分數衡量，其中較高數值代表較佳之 ESG 績效。

研究結果顯示，ESG 績效與智慧資本對企業績效具有正向且顯著之影響，無論以獲利能力或市場價值衡量皆然。相反地，槓桿程度對企業績效具有負向影響；而流動性雖呈現正向影響，但其顯著性並不一致。此外，研究亦發現，公司規模能強化 ESG 績效與智慧資本對企業績效之影響。

本研究結果顯示，有效的 ESG 實踐可透過較佳的企業風險管理、聲譽提升與營運效率改善，進而提升企業績效。本研究對企業管理者與監管機構具有實務貢獻，可作為推動 ESG 整合並將其作為企業永續經營策略之參考。此外，本研究亦具理論貢獻，有助於強化合法性理論、利害關係人理論與訊號理論在印尼資本市場情境下之解釋力。本研究亦豐富了開發中國家之 ESG 實證文獻，尤其是在印尼此一 ESG 風險基礎研究方法仍相對有限的情境中。進一步而言，本研究強調企業在策略決策過程中整合永續性與財務因素之重要性，使企業不僅關注短期績效，亦能兼顧長期永續發展。

關鍵詞： ESG；企業績效；智慧資本；槓桿；流動性。

ABSTRACT

This study aims to analyze the influence of Environmental, Social, and Governance (ESG) performance, intellectual capital, leverage, and liquidity on company performance, with firm size as a moderating variable. Company performance is measured using Return on Assets (ROA) as a short-term performance indicator and Tobin's Q as a long-term performance indicator. This study uses secondary data obtained from the financial statements of companies listed on the IDX ESG Leaders (IDXESGL) index for the period 2021–2023. A purposive sampling method was used to determine the research sample, resulting in 58 companies with a total of 174 panel data observations.

The analysis was conducted using panel data regression with a Fixed Effects Model (FEM) approach and Moderated Regression Analysis (MRA) to test the moderating role of firm size. ESG was measured using a transformed ESG Risk score, where higher values reflect better ESG performance.

The results of the study indicate that ESG performance and intellectual capital have a positive and significant impact on company performance, both in terms of profitability and market value. Conversely, leverage negatively impacts company performance, while liquidity shows a positive but not always significant effect. Furthermore, firm size has been shown to strengthen the influence of ESG and intellectual capital on company performance.

These findings suggest that effective ESG implementation can improve corporate performance through better business risk management, enhanced reputation, and operational efficiency. This research provides practical contributions to corporate management and regulators in promoting ESG integration as a business sustainability strategy, as well as theoretical contributions in strengthening legitimacy theory, stakeholder theory, and signaling theory in the context of the Indonesian capital market. This research also contributes to enriching the empirical literature in developing countries, particularly Indonesia, where ESG risk-based approaches are still relatively limited. Furthermore, this research emphasizes the importance of integrating sustainability and financial factors in corporate strategic decision-making, ensuring that companies focus not only on short-term performance but also on long-term sustainability.

Keywords: ESG; Firm Performance; Intellectual Capital; Leverage; Liquidity.

ACKNOWLEDGEMENTS

First of all, I would like to express my deepest gratitude to Allah SWT for His blessings, guidance, strength, and mercy, which enabled me to complete this thesis.

I would also like to extend my sincere gratitude to my beloved parents, Mama and Ayah, for their unconditional love, endless prayers, encouragement, and support throughout my academic journey. Their sacrifices and belief in me have been an invaluable source of strength.

My deepest appreciation goes to my advisor, Professor Lee Cheng-Kang, for his guidance, valuable suggestions, constructive feedback, and continuous support during the completion of this thesis. His academic direction has greatly helped me improve the quality of this research.

I would also like to express my heartfelt appreciation to Amin Yusuf Efendi, my beloved partner, for his patience, understanding, motivation, and emotional support throughout the thesis writing process. His support has been meaningful during every challenging stage of this journey.

Lastly, I would like to thank everyone who has contributed directly or indirectly to the completion of this thesis. May Allah SWT reward all the kindness, prayers, and support that have been given to me.

TABLE OF CONTENTS

摘要	iv
ABSTRACT	v
ACKNOWLEDGEMENTS	vi
LIST OF TABLES	x
LIST OF FIGURES	xi
CHAPTER 1 INTRODUCTIONS	1
1.1 Background	1
1.2 Formulation of the problem	6
1.3 Research purposes	6
1.4 Benefits of research	7
1.4.1 Practical Benefits	7
1.4.2 Theoretical Benefits	7
CHAPTER 2 LITERATURE REVIEW	8
2.1 Theoretical Basis and Definition of Variables	8
2.1.1 Legitimacy Theory	8
2.1.2 Stakeholder Theory	9
2.1.3 Signaling Theory	9
2.1.4 Indonesian Stock Exchange ESG Leader Index	10
2.1.5 Environmental, Social, Governance	11
2.1.6 Intellectual Capital	14
2.1.7 Leverage	14
2.1.8 Liquidity	15
2.1.9 Firm performance	15
2.2 Research Hypothesis	17
2.2.1 The influence of ESG on firm performance	17
2.2.2 The Influence of Intellectual Capital on Firm Performance	18
2.2.3 The Effect of Leverage on Firm Performance	19
2.2.4 The Influence of Liquidity on Firm Performance	19
2.2.5 The Effect of ESG on Firm Performance is Moderated by Firm Size	19

2.2.6 The Influence of Intellectual Capital on Firm performance Moderated by Firm Size	19
2.2.7 The Effect of Leverage on Firm Performance Moderated by Firm Size	20
2.2.8 The Effect of Liquidity on Firm Performance Moderated by Firm Size	20
2.3 Research Framework	20
CHAPTER 3 METHODOLOGY	22
3.1 Population and Sample	22
3.2 Data Types and Data Sources	23
3.3 Research Variables and Operational Definitions	23
3.2.1 Dependent Variable	23
3.2.2 Independent Variables	23
3.2.3 Moderation Variables	25
3.2.4 Control Variables	25
3.3 Data Analysis Methods	27
3.3.1 Classical Assumption Test	27
3.3.1.1 Normality Test	27
3.3.1.2 Heteroscedasticity Test	27
3.3.1.3 Autocorrelation Test	27
3.3.1.4 Multicollinearity Test	28
3.3.2 Hypothesis Testing	27
3.3.3 Coefficient of Determination	29
3.3.4 Moderated Regression Analysis (MRA)	29
3.3.5 Regression Model Specification	30
CHAPTER 4 RESULT	32
4.1 Descriptive Statistics	32
4.2 Correlation Test	33
4.3 Multicollinearity Test	34
4.4 Panel Model Selection (Hausman Test)	34
4.5 Fixed Effect + Moderation Regression Results	35
4.6 Robustness Check (Robust Standard Error)	37
4.8 Discussion	38
CHAPTER 5 CONCLUSION	42

5.1 Conclusion	42
5.2 Implications of Research Results	43
5.2.1 Theoretical Implications	43
5.2.2 Practical Implications	43
5.2.3 Policy Implications	44
5.3 Suggestions	44
REFERENCES	45
APPENDIX A	48

LIST OF TABLES

Table 3.1 Variable Measurement	26
Table 4.1 Descriptive Statistics	32
Table 4.2 Correlation Matrix	33
Table 4.3 Variance Inflation Factor (VIF)	34
Table 4.4 Hausman Test (FEM vs REM)	35
Table 4.5 ROA Regression Results (FEM + MRA)	35
Table 4.6 Tobin's Q Regression Results (FEM + MRA)	36
Table 4.7 Robustness Check (Summary)	37
Table A1 Research Data	48

LIST OF FIGURES

Figure 2.1 Research Thinking Framework	21
--	----

CHAPTER 1

INTRODUCTION

1.1 Background

Sustainable business can be achieved if the company can carry out economic aspects related to profit, social aspects related to people, and environmental aspects related to the planet (Persakis, 2024) and supported by the implementation of effective corporate governance (Demiraj et al., 2025). Sustainable business can be maintained if business activities do not damage natural resources and the environment so that the fulfillment of the needs of future generations can be maintained (Ho & Lu, 2025). In the current business context, business sustainability aims to meet economic, social, and environmental aspects by improving environmental, social, governance performance (Moodley, 2025). Environmental, social, governance performance shows the company's achievements in improving social and environmental aspects and implementing business ethics through effective governance (Mohammed Abubakr et al., 2026).

On the other hand, business activities carried out are too focused on economic performance and profit achievement, thus ignoring the impact of business on environmental damage (Elamer & Boulhaga, 2024), social, and environmental (Habib et al., 2025). This is not in line with the concept of sustainable business because business is too focused on current economic performance so that the sustainability of future performance will become a new problem. Several cases in Indonesia show that business activities have not yet carried out social and environmental responsibilities properly. The Ministry of Environment and Forestry reported the development of environmental damage cases until 2018, namely 462 companies involved in environmental damage cases (Persakis, 2024). Some of these cases include cases that occurred in the Freeport Indonesia company and the Lapindo Brantas company (Agustina et al., 2015). PT Freeport Indonesia dumped their business waste into the surrounding environment in Papua without permission from local residents and the government. The state suffered losses of USD 12.95 billion due to PT Freeport Indonesia's violations (Moussa & Elmarzouky, 2024). PT Lapindo Brantas harmed settlements and local residents in Sidoarjo, East Java due to oil drilling which caused a hot mud eruption (Agustina et al., 2015). The company's indifference to the impact of business activities on environmental damage that occurred was caused by weak environmental, social, and

governance performance (Zahrazova et al., 2025). In the context of emerging markets like Indonesia, ESG implementation is influenced not only by regulations but also by the level of corporate awareness and pressure from global investors. This leads to variations in the quality of ESG implementation across companies. Furthermore, a gap remains between administrative compliance with regulations and the substantive implementation of ESG in company operations. This situation demonstrates that ESG measurement cannot be based solely on disclosure; an approach capable of capturing the actual effectiveness of ESG implementation is needed.

The Indonesian government has made regulations related to the implementation of social and environmental responsibility, namely through Law no. 40 of 2007 concerning "Limited Liability Companies" and PP no. 47 of 2012 concerning "Social and Environmental Responsibility of Limited Liability Companies". These regulations require social and environmental responsibility programs by companies, but there are no more detailed programs and procedures for how social and environmental responsibility is carried out by companies. In 2017, the Financial Services Authority issued regulations related to business sustainability through POJK no. 51/POJK.03/2017 concerning "Implementation of Sustainable Finance for Service Financial Institutions, Issuers, and Public Companies". POJK Regulation no. 51/POJK.03/2017 specifically improves business sustainability through sustainable finance programs. Based on POJK no. 51/POJK.03/2017, sustainable finance programs regulate companies to maintain the sustainability of the company's business by improving economic, social, and environmental aspects through the implementation of effective environmental, social, governance.

The effective implementation of environmental, social, governance can provide benefits for companies to support firm performance. The implementation of environmental, social, governance can improve learning for companies to identify business risks and innovations, improve reputation to take advantage of business opportunities, improve management quality through transparency, maintain long-term business sustainability, and support financial performance (Gidage & Bhide, 2025).

The Ministry of Environment and Forestry (2017, 2019) reported an increase in the number of corporate innovations after carrying out environmental responsibilities of 150 (2015), 260 (2016), 401 (2017), 542 (2018), and 794 (2019). The Ministry of Environment and Forestry (2018, 2019) also reported that the implementation of innovation supported savings in corporate expenses of IDR 53.08 trillion (2017), IDR 287.34 trillion (2018), and

IDR 192.63 trillion (2019). Improving corporate reputation can increase business opportunities such as retaining quality employees, creating new markets for consumers and suppliers of goods that support the concept of green business, and reducing the cost of conflict with regulators and society (Sharpe et al., 2026; Gidage & Bhide, 2025; Habib et al., 2025). Business sustainability can also provide signals regarding the availability of cash flow and corporate profits (Mohammed Abubakr et al., 2026). Furthermore, the development of sustainable investment in recent years has also been a significant factor driving companies to improve their ESG performance. Institutional investors are increasingly considering non-financial aspects when assessing a company's prospects. Therefore, companies that effectively integrate ESG tend to have greater investment appeal than those that focus solely on traditional financial performance.

This study examines the influence of environmental, social, and governance performance in improving the financial performance of companies, especially companies whose shares are listed on the Indonesian Stock Exchange ESG Leader (IDXESGL) index. The IDXESGL index is a stock index published by the Indonesia Stock Exchange which contains shares of companies with the highest environmental, social, and governance performance. The assessment of environmental, social, and governance performance in IDXESGL uses ESG Risk which assesses how much risk to the company's economic value is influenced by environmental, social, and governance factors. The consideration of selecting companies listed on the IDXESGL index is because these companies are the focus for investors who support green business and the implementation and assessment of environmental, social, and governance performance.

Mohammed Abubakr et al. (2026), Zahrazova et al. (2025), Nguyen et al. (2022), Sahut and Pasquini-Descomps (2015), Velte (2017), and Aydoğmuş et al. (2022) found that environmental, social, and governance performance improves firm performance. In Indonesia, Tuti et al. (2025), Isma et al. (2025), and Pramesti et al. (2024) found that environmental, social, and governance factors improve company financial performance. On the other hand, Vieira et al. (2025), Sanoran (2026), and Tan and Wei (2024) did not find any effect of environmental, social, and governance factors on firm performance.

There are inconsistent research results, especially research in Indonesia regarding environmental, social, governance and financial performance. This is due to weaknesses in previous research in Indonesia. First, Tuti et al. (2025), Isma et al. (2025), Vieira et al. (2025), Tan and Wei (2024), and Sanoran (2026) measure environmental, social, governance factors

from the perspective of information disclosure in both the annual report and the sustainability report. Disclosure of environmental, social, governance information does not show performance that illustrates how much environmental, social, governance is effective and contributes to the company's business. Second, Pramesti et al. (2024) measures environmental, social, governance performance from ESG risk which looks at the effectiveness of environmental, social, governance in managing company risk, but only tests the effect on short-term performance, namely return on assets (ROA). In the context of signaling theory; environmental, social, governance provides signals regarding future business sustainability so that environmental, social, governance performance can also have an impact on future financial performance.

In contrast to previous studies in Indonesia such as Tuti et al. (2025), Isma et al. (2025), Vieira et al. (2025), Tan and Wei (2024), and Sanoran (2026), this study assesses environmental, social, governance performance not through disclosure, but through the magnitude of environmental, social, governance performance values. This study assesses environmental, social, governance performance through the effectiveness of environmental, social, governance implementation in supporting the company's business risk management. Based on the results of previous studies (Binesh et al., 2025; Tran & Nguyen, 2026; Moussa & Elmarzouky, 2024) which found that effective risk management can drive the company's financial performance, this study argues that the ability to implement environmental, social, governance that can reduce the company's business risk can encourage the company to improve the company's financial performance. This study assesses the ability to implement environmental, social, governance in supporting the company's business risk management through low ESG Risk through the evaluation of the IDXESGL index by the Indonesia Stock Exchange. The use of ESG risk measures in measuring environmental, social, and governance performance has been carried out by Pramesti et al. (2024).

Also different from previous studies in Indonesia such as Pramesti et al. (2024) which only tested short-term performance through ROA, this study assesses short-term (ROA) and long-term (Tobin's Q) financial performance. ROA measures performance in generating current period profits while Tobin's Q measures performance through the perspective of investors who assess the company's future prospects (Mohammed Abubakr et al., 2026). In addition to measurement weaknesses, previous research has tended to under-consider the role of moderating variables, particularly firm size, in strengthening or weakening the relationship between ESG and corporate performance. Yet, the resource capacities of large

and small companies differ significantly in implementing sustainability strategies. Therefore, this study attempts to address this gap by incorporating firm size as a moderating variable, which is expected to provide a deeper understanding of this relationship.

This study aims to obtain empirical evidence regarding the influence of environmental, social, and governance performance on firm performance. This study measures the company's financial performance from the aspects of return on assets (ROA), ROE, and Tobin's Q. ROA is an assessment of the ability of asset management to generate profits (Mohammed Abubakr et al., 2026). Tobin's Q measures how much the market value of a company's assets is based on market participants' assessment of the book value of assets (Mohammed Abubakr et al., 2026). This study follows the suggestions of Zahrazova et al. (2025) who looked at the impact of social and environmental responsibility from ROA which looked at the company's short-term performance in assessing the effectiveness of management using assets to generate current profits, as well as Tobin's Q which looked at the company's long-term performance in assessing the perspective of investors who evaluated the company's future performance prospects.

The background explanation shows some core issues of this research. First, the implementation of environmental, social, governance is important in the framework of business sustainability through economic, social, and environmental aspects. Second, there are cases of environmental damage by companies in Indonesia. Third, there is regulation POJK no. 51/POJK.03/2017 which requires the implementation of sustainability programs through environmental, social, governance aspects.

Fourth, the implementation of environmental, social, governance provides benefits for companies that support the improvement of the company's financial performance. Fifth, previous research in Indonesia has weaknesses regarding the measurement of environmental, social, governance performance and financial performance. Fifth, in Indonesia, the IDXESGL index provides an assessment of the effectiveness of environmental, social, governance on the management of company business risks. Based on the explanation of the background, this study examines the "Effect of Environmental, Social, Governance Performance on the Performance of Companies Listed on the Indonesian Stock Exchange ESG Leader Index".

1.2 Formulation of the problem

Based on the explanation of the background , this study formulates the problem as follows,

1. Does environmental, social and governance performance affect firm performance?
2. Does intellectual capital affect firm performance?
3. Does leverage affect firm performance?
4. Does liquidity affect firm performance?
5. Does environmental, social, and governance performance affect firm performance moderated by firm size?
6. Does intellectual capital influence firm performance moderated by firm size?
7. Does leverage affect firm performance moderated by firm size?
8. Does liquidity affect firm performance moderated by firm size?

1.3 Research purposes

The objectives of this research are as follows,

1. Obtaining empirical evidence regarding the influence of environmental, social and governance performance on company operational performance.
2. Obtaining empirical evidence regarding the influence of intellectual capital on firm performance.
3. Obtaining empirical evidence regarding the effect of leverage on firm performance.
4. Obtaining empirical evidence regarding the effect of liquidity on firm performance.
5. Obtaining empirical evidence regarding the influence of environmental, social and governance performance on company operational performance moderated by firm size.
6. Obtaining empirical evidence regarding the influence of intellectual capital on firm performance moderated by firm size.
7. Obtaining empirical evidence regarding the effect of leverage on firm performance moderated by firm size.
8. Obtaining empirical evidence regarding the effect of liquidity on firm performance moderated by firm size.

1.4 Benefits of research

The benefits of this research are as follows.

1.4.1 Practical Benefits

Companies can improve environmental, social, governance performance to support the improvement of the company's financial performance. The results of this study can also be input for regulators to evaluate environmental, social, governance performance for all companies, not only companies listed on the IDXESGL index. The results of this study are also expected to be investment considerations for capital market players in investing funds in stocks.

A company that supports business sustainability by improving environmental, social and governance performance.

1.4.2 Theoretical Benefits

The results of this study are additional literature, especially those related to the concept of business sustainability and social and environmental responsibility, especially regarding the influence of environmental, social and governance performance on financial performance.



CHAPTER 2

LITERATURE REVIEW

2.1 Theoretical Basis and Definition of Variables

This study explains the theories that are the basis of the research. The theories used consist of legitimacy theory, stakeholder theory, and signaling theory. This study also explains the concept of the Indonesian Stock Exchange ESG index, the concept of environmental, social, governance, and the concept of financial performance which are the research variables.

2.1.1 Legitimacy Theory

Legitimacy theory explains why companies tend to consider social and environmental aspects when conducting business processes, including environmental, social, and governance aspects. Suchman (1995) explains the element of legitimacy as a general assumption and perception that explains that a company's business activities are part of the overall system of values, norms, and social beliefs. Companies will improve and maintain the legitimacy of their business activities by implementing environmental, social, and governance aspects (Kilian & Hennigs, 2014). Companies implement environmental, social, and governance aspects with the motivation of legitimacy for the company's business activities (Faisal et al., 2012).

A company is an organization built on public trust in the company (O'Dwyer et al., 2011) and tends to maintain the company's position in society to ensure the continuity of the company's business (Erin et al., 2022). In the context of environmental, social, governance, Maroun et al. (2014) explained that social and environmental responsibility under effective governance is part of the legitimacy process that is designed to provide signals about the company's reporting ability and governance system to respond to stakeholder interests. Although the implementation of environmental, social, governance is made in order to fulfill regulatory obligations, legitimacy must still be built through social and environmental responsibility and business sustainability to attract the attention of investors and other stakeholders (Maroun et al., 2014). From the perspective of legitimacy theory, the environmental, social, governance aspects aim to legitimize the company's activities based on the expectations of various stakeholders. The company maintains its legitimacy by giving

signals to stakeholders that the company's behavior is in accordance with stakeholder expectations (Suchman, 1995). In the context of this research, legitimacy is viewed not only as a company's effort to gain social acceptance but also as a strategy to maintain long-term business sustainability. ESG implementation is a key instrument in building this legitimacy, especially amidst increasing demands for transparency and accountability from various parties.

2.1.2 Stakeholder Theory

Stakeholder theory is a theory that explains motivation and determines sustainability reporting practices. Freeman and Dmytriiev (2017) explain stakeholder theory as a theory that emphasizes the position of stakeholders as individuals or groups involved or directly and indirectly affected by a company's business activities, including social and environmental impacts. Stakeholders include formal and informal groups or organizations and are the main elements of a company's external scope that can have a good or bad effect on the company (Beasley & Salterio, 2001). The theory *Stakeholder focus* on fulfilling stakeholder interests and expectations and managing conflicts of interest between stakeholders (Ngatia, 2014). Spence and Rinaldi (2014) argue that the implementation of environmental, social, governance must be based on stakeholder theory because firm performance should contribute to meeting the needs of all stakeholders. Companies must consider stakeholder expectations and interests to support the company's business interests (Erin et al., 2022). Thus, corporate success is no longer measured solely by its ability to generate profits, but also by its ability to create value for all stakeholders. This makes ESG a relevant strategic approach for bridging economic and social interests.

2.1.3 Signaling Theory

Signaling theory explains the actions of companies to provide a signal of information to external parties of the company regarding private information that is not yet known by external parties (Kirmani and Rao 2000). The purpose of giving signals is to provide information about the quality of the company compared to other companies. In the context of implementing environmental, social, governance, companies aim to provide signals about environmental, social, governance benefits and communicate business sustainability in the future (King et al., 2005; Moratis, 2018).

Lee et al. (2022) explained that signaling activities regarding environmental, social,

and governance aspects are also a marketing tool for companies to suppliers, employees, consumers, and other parties. By implementing environmental, social, and governance aspects, companies strive to gain credibility and a good reputation in the eyes of suppliers, employees, consumers, and investors, thereby supporting the company's business activities (Lee et al., 2022). In signaling theory, environmental, social, and governance assessments focus on the perspectives and assessments of external parties. In this study, ESG is viewed as a non-financial signal that plays a crucial role in reducing information asymmetry between management and investors. The better a company's ESG performance, the stronger the positive signal it sends to the market regarding the company's future prospects.

2.1.4 Indonesian Stock Exchange ESG Leader Index

The Indonesian Stock Exchange ESG Leader Index (IDXESGL) is a new index on the Indonesia Stock Exchange issued on December 14, 2020. The IDXESGL index is a thematic index on the Indonesia Stock Exchange that has a theme and category for assessing environmental, social, and governance aspects. The IDXESGL index is an index containing shares of companies that have performed well in environmental, social, and governance aspects; the company is not a company that has significant controversy; and companies with satisfactory stock transaction liquidity and financial performance through an evaluation process by a third party, namely Sustainalytics (Indonesia Stock Exchange, 2021).

Environmental, social, governance performance in the IDXESGL index assessment uses ESG risk. ESG risk measures the extent to which the risk to the company's economic value is influenced by environmental, social, and governance factors (Standard Chartered, 2021). The lower the ESG risk, the higher the environmental, social, and governance ability to reduce and manage the company's business risks.

ESG risk assessment begins with determining the scope of business risk influenced by the implementation of environmental, social, governance. These risks consist of business risks that can be managed by the company and risks that cannot be managed by the company. Risks that can be managed by the company consist of risks that have been managed and risks that have not been managed. Risks that have been managed indicate the benefits of implementing environmental, social, governance for the company to help manage and reduce business risks. On the other hand, risks that cannot and have not been managed properly by the company indicate that the implementation of environmental, social, governance has failed to reduce the company's business risk. The failure of implementing environmental,

social, governance in reducing the company's business risk reflects ESG risk. The higher the ESG risk, the lower the effectiveness of implementing environmental, social, governance, while the lower the ESG risk, the higher the effectiveness of implementing environmental, social, governance. This study argues that the effectiveness of implementing environmental, social, governance can help companies improve their financial performance.

2.1.5 Environmental, Social, Governance

Environmental, social, governance aspects are aspects that must be fulfilled in supporting economic, social and environmental performance to achieve business sustainability goals (Shakil, 2021). Baier et al. (2020) also explained that environmental, social, governance aspects are the foundation for implementing corporate social responsibility and sustainable business. The implementation of environmental, social, governance aspects is a single activity. Environmental, social, governance is defined as the activity of fulfilling social and environmental responsibilities and implementing business ethics under effective supervision (governance) (Kim & Li, 2021). Each aspect has components that are interrelated with each other. The components of the environmental, social, governance aspects are as follows (Baier et al., 2020).

2.1.5.1 Governance Aspects

Governance aspects include corporate governance mechanisms, audit and control, corporate board structure, management remuneration, shareholder rights, implementation of transparency, human resource fulfillment, business ethics, fraud and corruption, political impact, accountability, whistle-blowing system, sustainability management and reporting, disclosure and reporting, stakeholder engagement, compliance with regulations and regulators, governance of sustainability issues.

2.1.5.2 Environmental Aspects

Environmental aspects include ecosystem conservation, access to land, biodiversity management, water use and water resources, climate change, biofuel use, climate change strategies, emissions management and reporting, environmental management, environmental standards, pollution control, product responsibility, waste and recycling, and environmental supply chain standards.

2.1.5.3 Social Aspect

Social aspects include public health, access to treatment, HIV and AIDS, nutritional fulfillment, product safety, community, charity programs, education, employees, human rights, community relations, freedom of expression, labor standards, diversity, health and safety, labor organizations, labor supply chain standards.

There are several benefits that can be obtained by the company if the company implements environmental, social, governance. The Association of Chartered Certified Accountants (2013) explains the benefits of implementing/including environmental, social, governance reporting as follows.

1. The application of environmental, social, governance can trigger learning and innovation by companies. The application of environmental, social, governance allows companies to identify and evaluate inefficient activities, business risks, and new innovations for sustainable management and business development.
2. The implementation of environmental, social, governance helps companies improve their reputation and competitive advantage. The implementation of environmental, social, governance improves the company's reputation for employees, investors, and other stakeholders, thus helping the company to build cooperation and engage with new businesses or markets. A good corporate reputation maintains employee loyalty, attracts investors, and improves stakeholder relations. In the future, the implementation of environmental, social, governance can be used as a complement to non-financial requirements for entering new markets, for example as a requirement for running an initial public offering (IPO) process.
3. The implementation of environmental, social, governance helps companies to improve and identify business opportunities. This happens because there is an opportunity to attract new employees with high competence, attract investors, and build good communication with the government, regulators, non-profit organizations and the public sector, and local communities.
4. The implementation of environmental, social, governance can increase the transparency of the company. This can increase the trust of employees, stakeholders, and investors. Transparency can also improve the quality of management and the value of the company.
5. The implementation of environmental, social, governance can be used as a medium for assessment by stakeholders. The implementation of environmental, social,

governance is an indicator of assessment by stakeholders on how the company is managed.

6. The implementation of environmental, social, governance can be used as an assessment of business risks and opportunities. Environmental, social, governance performance is used as an indicator of business resilience and sustainability, including non-financial risks (social and environmental risks).
7. The implementation of environmental, social, and governance can maintain communication and relationships with stakeholders.
8. The application of environmental, social and governance can be used as additional information that can strengthen the evaluation of financial information.

In Indonesia, the implementation of environmental, social, governance is included in the sustainability program regulated in POJK regulation no. 51/POJK.03/2017 concerning "Implementation of Sustainable Finance for Financial Services Institutions, Issuers, and Public Companies". Companies are required to implement a sustainable finance program by conducting and reporting business activities that support economic, social and environmental aspects. In the implementation of sustainable finance, POJK no. 51/POJK.03/2017 provides implementation procedures, one of which is through improving environmental, social, governance performance. The implementation of POJK regulation no. 51/POJK.03/2017 is required between 2019-2023 for the implementation of sustainability activities and the 2020-2024 financial year to implement sustainability reports. To support this implementation, the Indonesia Stock Exchange published a special index that collects shares of companies with the best environmental, social, governance performance, namely the IDXESGL index in 2020.

Environmental, social, and governance research continues to develop, especially regarding the impact of environmental, social, and governance on corporate financial performance. Friede et al. (2015) reported that there has been empirical research as many as 22,000 studies on environmental, social, governance and corporate performance between 1975-2015. Whelan et al. (2021) also reported that there were more than 1,000 studies on environmental, social, governance and corporate performance between 2015-2020. On the other hand, the large number of studies on environmental, social, governance and corporate performance actually creates inconsistent findings on the relationship between environmental, social, governance and corporate performance (Whelan et al., 2021). This

occurs because previous studies still incorrectly identify performance, significant activities, and environmental, social, governance disclosures (Whelan et al., 2021). However, ESG implementation also faces various challenges, such as high implementation costs, limited human resources, and a lack of uniform measurement standards. Therefore, the effectiveness of ESG in improving company performance remains a matter of debate in the empirical literature.

2.1.6 Intellectual Capital

Intellectual capital (IC) is an intangible asset consisting of three main components: human capital, structural capital, and relational capital. Human capital includes the knowledge, skills, and competencies of employees that contribute directly to the creation of value for the company (Edvinsson & Malone, 1997). Structural capital includes the organization's infrastructure, processes, and databases that support human capital, while relational capital refers to the company's relationships with customers, suppliers, and other stakeholders (Stewart, 1997). These three components play a role in increasing competitive advantage and organizational performance.

IC is considered as a key driver of innovation and economic growth in a knowledge-based economy. Previous studies have shown that companies with good IC management tend to have higher productivity and profitability (Pulic, 2000). In the context of SMEs, IC is also an important factor in increasing competitiveness in a dynamic market (Bontis, 1998). Therefore, IC measurement and management have become increasingly attractive research areas in the accounting and strategic management literature. Various models have been developed to measure IC, one of which is the Value Added Intellectual Coefficient (VAIC™) introduced by Pulic (2000). This model allows companies to evaluate the contribution of intellectual capital to the creation of added value for the company. In addition, IC measurement can also be used as a diagnostic tool to assess the effectiveness of innovation strategies and human resource development (Chen et al., 2005).

2.1.7 Leverage

Financial leverage refers to the use of debt in a firm's capital structure to increase shareholder returns. This concept is based on the capital structure theory proposed by Modigliani and Miller (1958), which states that capital structure does not affect firm value under perfect market conditions. However, in practice, leverage can affect a firm's financial risk and profitability.

High leverage levels can provide benefits in the form of tax savings through deductible debt interest (tax shield), but also increase the risk of bankruptcy (Jensen &

Meckling, 1976). Empirical research shows that the relationship between leverage and financial performance tends to be non-linear, where moderate leverage can improve performance, but excessive leverage can harm the company (Myers, 1977). Therefore, optimal leverage management is very important to maintain a balance between risk and return.

In the context of MSMEs, leverage is often an important tool to obtain working capital and support business expansion. However, access to debt is often limited by repayment capacity and credit risk. Research shows that appropriate debt structure can improve operational efficiency and financial stability of MSMEs, but also requires careful risk management (Titman & Wessels, 1988).

2.1.8 Liquidity

Liquidity reflects a company's ability to meet its short-term obligations using current assets. Good liquidity indicates that a company has the financial flexibility to manage daily operations and deal with emergencies (Gitman & Zutter, 2012). Liquidity ratios, such as the current ratio and quick ratio, are often used to assess a company's liquidity level.

However, too much liquidity can indicate inefficient asset allocation, such as cash accumulation that is not productively invested (Van Horne & Wachowicz, 2005). Conversely, low liquidity can increase default risk and reduce investor confidence. Previous research has found that liquidity acts as a buffer against earnings volatility and market uncertainty, especially in financial crisis situations (Brealey et al., 2011).

In MSMEs, liquidity is often a major challenge due to limited resources and access to external funding. Effective liquidity management strategies, such as managing receivables and inventory, can help MSMEs maintain operational continuity and increase profitability (Ross et al., 2016). Therefore, liquidity is one of the key indicators in assessing the financial health and sustainability of MSME businesses.

2.1.9 Firm performance

Financial performance is a company's achievement when carrying out business activities as measured by financial ratios (Yoon & Chung, 2018). Tarigan and Samuel (2015) explain several financial ratios in describing the company's performance conditions such as asset management (inventory turnover, receivables, working capital, fixed assets, and total assets), profitability (profit margin, ROA, and ROE), leverage (debt equity ratio and cash

coverage), liquidity (consisting of current ratio, quick ratio, and cash ratio), and stock market aspects (Tobin's q).

This study focuses on short-term financial performance represented by ROA and long-term financial performance represented by Tobin's q. Short-term financial performance shows the company's performance in the current period. ROA is chosen as short-term performance because it measures the ability to use assets to generate profits in the current year. Long-term financial performance shows the company's performance in the future. Tobin's q is chosen as long-term performance because it measures the company's potential future performance through investor perceptions of the company's future prospects.

ROE describes how much profit is generated in the use of assets. An increase in ROA indicates an increase in the company's effectiveness in using assets so that the company makes a profit. Tobin's q is a measure of market performance that measures the market value of assets against book value. An increase in Tobin's q indicates that the market assesses the company as being in good condition. Tobin's q that is more than 1 indicates that the market assesses the company's assets above their book value, while Tobin's q below 1 indicates that the market assesses the company's assets below their book value.

There are several considerations in selecting ROA. First, previous studies related to the implementation of environmental, social, governance also use ROA as a company's financial performance such as research by Kim and Li (2021), Hwang et al. (2021), Yoo and Managi (2022), Husada and Handayani (2021), Tarigan and Semuel (2015), and Juliandara et al. (2021). Second, profit is one of the main goals of business so that the use of ROA is often used as a measure of firm performance (Henisz et al., 2019). Third, the benefits of environmental, social, governance partly affect the components of profit calculation such as new market share, employee productivity, and reputation that increase sales or revenue, as well as innovation, compliance, and transparency that can reduce costs (Henisz et al., 2019). Fourth, ROA shows the effectiveness of the company in using available resources to generate profits after benefiting from the implementation of environmental, social, governance (Yoon & Chung, 2018). The calculation of ROA can be seen as follows (Yoon & Chung, 2018).

This study also has several considerations in selecting Tobin's q. First, there are benefits of implementing environmental, social, governance to attract investors (The Association of Chartered Certified Accountants, 2013) so that market performance can be used as a measure of firm performance. Second, one of the foundations of implementing environmental, social, governance is signaling theory so that a market response to the signal

is needed. Third, the implementation of environmental, social, governance is the foundation for future business sustainability so that long-term performance needs to be assessed through the investor's perspective on the company's future prospects (Yoon & Chung, 2018). Fourth, previous studies such as Tarigan and Samuel (2015), Yoon and Chung (2018), and Yoo and Managi (2022) also use Tobin's q as the company's financial performance in the context of implementing environmental, social, governance. The calculation of Tobin's q is calculated as follows (Yoon & Chung, 2018).

$$Tobin's\ q = \frac{Market\ value\ of\ equity + Book\ value\ of\ debt}{Book\ value\ of\ assets}$$

2.2 Research Hypothesis

2.2.1 The influence of ESG on firm performance

Effective implementation of environmental, social, governance can benefit the company. The implementation of environmental, social, governance can trigger the development and innovation of the company. The company can conduct research and development of environmentally friendly products and create potential new market shares for consumers who are oriented towards green products (Broadstock et al., 2020). The development of new market shares will increase the company's revenue growth. The development of environmentally friendly products can be carried out through environmental aspects (by implementing product responsibility) and social aspects (by implementing product safety) (Baier et al., 2020).

Another innovation is developing certain technologies to increase the productivity of human resources in the production process of goods and services and reducing machines that can increase pollution (Zhang et al., 2020). Increased productivity can increase sales and reduced machine costs can increase efficiency (Broadstock et al., 2020). Production innovation can be through environmental aspects (by implementing water, fuel, pollution and waste management) and social aspects (by implementing employee and workforce education) (Baier et al., 2020)

Based on stakeholder theory, the application of environmental, social, governance is a tool to fulfill stakeholder expectations and desires. Innovation is one of the company's responses in fulfilling consumer expectations and desires to obtain environmentally friendly products, as well as the expectations and desires of the community not to suffer from

pollution from the company's production activities. The application of environmental, social, governance can also improve the company's reputation. The company's reputation is useful for retaining and recruiting quality and productive employees (Sun & Yu, 2015) so that it can increase the company's productivity (The Association of Chartered Certified Accountants, 2013). The benefits of improving reputation can be done through governance aspects (implementation of business ethics, transparency, and regulation) and social aspects (implementation of public relations, employee management, and employment systems) (Baier et al., 2020). Based on signaling theory, the company gives a signal regarding the company's good reputation. In this research study, the effectiveness of the company's resources in obtaining current period profits is described by ROA. ROA is used to measure short-term financial performance. The study uses ROA in examining the relationship between the application of environmental, social, governance and ROA. On the one hand, Yoo and Managi (2022), Kim and Li (2021), Hwang et al. (2021), Safriani and Utomo (2020), Husada and Handayani (2021), and Priandhana (2022) found that environmental, social, and governance performance increased ROA. On the other hand, Juliandara et al. (2021), Yoon and Chung (2018), and Tarigan and Samuel (2015) did not find an effect of environmental, social, and governance performance on ROA. This explanation indicates H1 as follows.

H1: Environmental, social, and governance performance has an effect on firm performance.

2.2.2 The Influence of Intellectual Capital on Firm Performance

Intellectual capital (IC) plays an important role in driving corporate performance. IC components such as human capital, structural capital, and relational capital support innovation and competitive advantage (Chen et al., 2005). A study by Pulic (2000) showed that companies with effective IC management tend to have higher productivity and profitability.

IC helps companies create added value through the management of intangible assets. Quality human capital, supported by good infrastructure and external relations, can improve operational efficiency and market competitiveness (Bontis, 1998). In the context of a knowledge-based economy, IC becomes a key factor in achieving long-term growth.

H2: Intellectual capital has a positive effect on firm performance.

2.2.3 The Effect of Leverage on Firm Performance

Financial leverage can affect firm performance through tax shield effects and increased equity returns (Jensen & Meckling, 1976). However, high leverage can also increase the risk of bankruptcy, which has a negative impact on performance (Myers, 1977).

Empirical research shows that the relationship between leverage and financial performance is non-linear, with moderate leverage providing optimal returns (Titman & Wessels, 1988). In stable industries, leverage tends to have a positive impact because it encourages business expansion and investment.

H3: Leverage has a positive effect on firm performance.

2.2.4 The Influence of Liquidity on Firm performance

Liquidity reflects a company's ability to meet short-term obligations. Adequate liquidity can increase operational flexibility and reduce financial risk (Gitman & Zutter, 2012). However, excessive liquidity can reflect inefficient resource allocation (Van Horne & Wachowicz, 2005).

Studies show that optimal liquidity contributes to financial stability and profitability of companies, especially in the face of market uncertainty (Brealey et al., 2011). In certain sectors, liquidity is also a key indicator in attracting investment.

H4: Liquidity has a positive effect on firm performance.

2.2.5 The Effect of ESG on Firm performance is Moderated by Firm Size

Environmental, Social, and Governance (ESG) factors are a major concern in improving firm performance. Companies with high ESG scores tend to attract investor attention and create market confidence (Friede et al., 2015). However, the effect of ESG on performance can vary depending on company size.

Larger firms have greater resources to integrate ESG practices than smaller firms. Firm size also moderates the relationship between ESG and performance through the effectiveness of sustainability strategy implementation (Eccles et al., 2014).

H5: Firm size moderates the effect of ESG on firm performance.

2.2.6 The Influence of Intellectual Capital on Firm performance Moderated by Firm Size

Company size can strengthen the influence of IC on performance. Large companies

have better capacity to manage and optimize IC components such as human and relational capital (Bontis, 1998).

A study by Chen et al. (2005) showed that the influence of IC on performance is more significant in large companies due to adequate infrastructure and resource support. In addition, large companies tend to be more efficient in distributing IC benefits.

H6: Firm size moderates the influence of intellectual capital on firm performance.

2.2.7 The Effect of Leverage on Firm Performance Moderated by Firm Size

Firm size has the potential to moderate the relationship between leverage and performance. Large firms have higher bargaining power over creditors, so they can access debt with lower interest rates (Titman & Wessels, 1988).

Research shows that leverage has a more significant impact on large companies due to better cash flow stability (Myers, 1977). Conversely, in small companies, high leverage can increase financial risk.

H7: Firm size moderates the effect of leverage on firm performance.

2.2.8 The Effect of Liquidity on Firm Performance Moderated by Firm Size

Liquidity has different impacts on firm performance based on firm size. Larger firms typically have broader access to funding sources, so the relationship between liquidity and performance may differ compared to smaller firms (Brealey et al., 2011).

In large companies, high liquidity tends to support operational stability without sacrificing efficiency. Meanwhile, small companies are more vulnerable to low liquidity due to resource constraints.

H8: Firm size moderates the effect of liquidity on firm performance.

2.3 Research Framework

Based on the theoretical foundation and hypothesis development, this study examines the effects of ESG performance, intellectual capital, leverage, and liquidity on firm performance. Firm performance is measured using two indicators, namely ROA and Tobin's Q. ROA represents accounting-based short-term performance, while Tobin's Q represents market-based long-term performance.

This study also examines the moderating role of firm size. Firm size is expected to influence the strength of the relationship between ESG performance, intellectual capital,

leverage, liquidity, and firm performance. In addition, firm age and sales growth are included as control variables to reduce omitted variable bias and improve the robustness of the model. The relationship among the variables is presented in Figure 2.1.

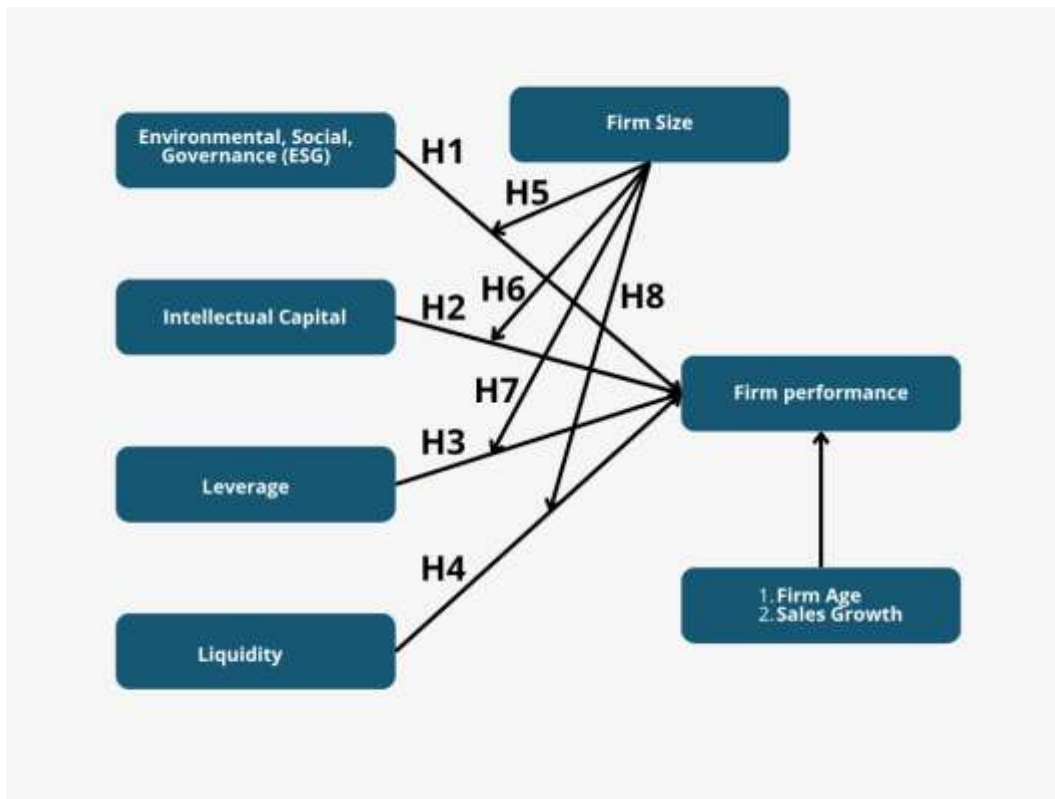


Figure 2.1 Research Thinking Framework

Figure 2.1 shows that ESG performance, intellectual capital, leverage, and liquidity are independent variables that are expected to influence firm performance. Firm size is positioned as a moderating variable that may strengthen or weaken the effect of each independent variable on firm performance. Firm age and sales growth are included as control variables because firm maturity and business growth may also influence firm performance. Therefore, the framework reflects the integrated relationship among sustainability performance, intangible resources, financial structure, financial flexibility, firm size, and firm performance.

CHAPTER 3

METHODOLOGY

3.1 Population and Sample

The population of this study consists of all firms listed on the Indonesia Stock Exchange (IDX) during the period 2021–2023. The Indonesia Stock Exchange was selected because it provides publicly available and reliable financial and ESG-related data required for this study. The research sample consists of firms selected from the population using purposive sampling method. Purposive sampling is applied to select firms that meet specific criteria relevant to the research objectives. Purposive sampling refers to the selection of samples based on specific criteria to ensure the relevance and availability of data required for the analysis. The criteria that are appropriate to the discussion and topic of this research are explained as follows.

1. The sample consists of firms listed in the IDX ESG Leaders Index (IDXESGL). This study considers the IDXESGL index because it provides an assessment of environmental, social, and governance performance through ESG risk measurements. The IDXESGL index is also an index that provides a list of stocks for investors with a preference for environmental, social, and governance activities carried out by the company.
2. The firm has positive equity value to ensure financial viability and meaningful financial performance measurement. This consideration is based on the assessment of the company's future performance prospects by investors in the capital market. Negative equity indicates no investor contribution to the company's funding because the company's funding is entirely from creditors through debt financing.
3. The firm has complete financial and ESG data during the period 2021–2023. The selection of the data period is based on consideration of the IDXESGL index which was just launched in 2020.

Based on the IDXESGL index, there were 88 firms listed during the observation period. However, after applying the sample selection criteria, only 58 firms had complete ESG and financial data for the period 2021–2023. Therefore, the final sample consists of 58 firms that meet all the sample selection criteria. Since this study uses panel data covering

three years, the total number of firm-year observations is 174 observations. The selection of the 2021–2023 research period also took into account the stability of the IDXESGL index implementation following its launch in 2020, thus ensuring the data used is more representative in depicting the ESG conditions of companies in Indonesia.

3.2 Data Types and Data Sources

The type of data is secondary data. Secondary data is data that has been collected and provided by other parties. Data is taken from data sources in the form of company financial reports and can be accessed on the page www.idx.co.id. Documentation techniques are used as data collection techniques. Documentation techniques refer to collection techniques that originate from official documents prepared by the company and then published in the form of financial reports by the Indonesia Stock Exchange. Financial reports include annual financial reports and quarterly financial reports because they adjust the IDXESGL index issuance period. This study uses ESG risk, total assets, total liabilities, net income, stock market price, and outstanding shares. The use of secondary data in this study was chosen because it offers a higher level of objectivity and allows for analysis across a broader sample. However, the use of secondary data also has limitations, such as dependence on the availability and quality of data published by the company.

3.3 Research Variables and Operational Definitions

The research variables consist of independent variables, dependent variables, and control variables. Each variable is measured through a predetermined proxy.

3.3.1 Dependent Variable

The dependent variable is firm performance. Financial performance consists of short-term financial performance and long-term financial performance. ROA is used as a measure of operational performance, ROE is used as a measure of financial performance, while market performance uses Tobin's Q.

3.3.2 Independent Variables

Environmental, social, governance performance is an independent variable. Environmental, social, governance performance shows the effectiveness of the company in

implementing environmental, social, governance aspects in the business it runs. Environmental, social, governance performance is measured by the ESG risk value provided in the IDXESGL index. ESG risk shows the extent of the risk to the company's economic value caused by environmental, social, governance performance (Standard Chartered, 2021). ESG performance is measured using transformed ESG Risk score by multiplying ESG Risk by (-1), so that higher values indicate better ESG performance. Higher ESG risk indicates lower environmental, social, governance performance, while lower ESG risk indicates higher environmental, social, governance performance (Priandhana, 2022). ESG risk is the opposite measure of environmental, social, governance performance. The next independent variables are intellectual capital, leverage, and liquidity.

Intellectual capital (IC) is an intangible asset that reflects the knowledge resources owned by an organization to create added value and competitive advantage. IC consists of three main components: human capital, structural capital, and relational capital (Edvinsson & Malone, 1997). Human capital includes employee competencies, skills, and experience; structural capital includes the systems, processes, and organizational infrastructure that support human capital; while relational capital refers to the company's relationships with customers, partners, and other stakeholders (Bontis, 1998). Effective IC management can help companies improve innovation, operational efficiency, and overall financial performance.

Leverage is the use of external funding sources in a company's capital structure to finance operations or investment activities. Leverage can be measured through the ratio of debt to equity or total assets, which reflects the extent to which a company relies on debt to fund its business activities (Modigliani & Miller, 1958). In practice, leverage can provide benefits in the form of tax savings through deductible interest (tax shield), but also increases financial risk, especially in unstable market conditions (Jensen & Meckling, 1976). Therefore, optimal leverage management is needed to achieve a balance between risk and profit, and support the sustainability of firm performance.

Liquidity refers to a company's ability to meet its short-term obligations using current assets such as cash, receivables, and inventory. Liquidity ratios, such as the current ratio and quick ratio, are often used to measure the extent to which a company has financial flexibility in running its operations (Gitman & Zutter, 2012). Adequate liquidity reflects good financial management, but too much liquidity can indicate inefficiency in the use of assets, while too little liquidity increases the risk of default. In a business context, liquidity plays an important

role in maintaining the stability of day-to-day operations and supporting the company's resilience to changing market conditions.

3.3.3 Moderation Variables

The moderating variable in this study is firm size. Firm size refers to the scale of business operations which is usually measured through indicators such as total assets, revenue, number of employees, or market value of the company. Company size reflects the company's ability to manage resources, face risks, and attract investors (Rajan & Zingales, 1995). Large companies tend to have better access to external funding, more sophisticated infrastructure, and greater market influence than small companies. Company size is also often used as a moderating variable in research, because it can affect the relationship between independent variables and firm performance.

3.3.4 Control Variables

Control variables are included in this study to reduce potential omitted variable bias and to improve the robustness of the regression model in explaining firm performance. Control variables are factors that may influence firm performance but are not the main focus of the study. This study uses firm age and sales growth as control variables. Firm age reflects the firm's experience, operational stability, and accumulated knowledge, which may contribute to better performance. Older firms tend to have more established operational systems, stronger market positions, and better risk management capabilities compared to younger firms. Firm age is measured as the number of years since the firm was listed on the Indonesia Stock Exchange.

Sales growth is included as a control variable because it reflects the firm's business expansion and market demand, which may directly affect financial performance. Firms with higher sales growth tend to have better operational performance, increased profitability, and stronger market competitiveness. Sales growth is measured as the annual percentage change in total sales compared to the previous year. Including these control variables helps ensure that the estimated relationship between ESG, intellectual capital, leverage, liquidity, firm size, and firm performance is not biased by firm-specific characteristics related to firm maturity and growth opportunities. To ensure clarity and consistency in operationalizing the research variables, Table 3.1 summarizes the measurement of each variable used in this study.

Table 3.1 Variable Measurement

No	Variable Name	Definition	Measurement	Source
1	Environmental, Social, and Governance (ESG)	Sustainability dimensions that include environmental, social and governance aspects that are integrated into corporate strategy.	ESG scores published by ESG rating agencies (eg, Refinitiv or Bloomberg ESG Score).	Friede et al. (2019); Khan et al. (2022).
2	Intellectual Capital	Intangible assets consisting of human capital, structural capital, and relational capital to create added value.	Value Added Intellectual Coefficient (VAIC): $VAIC = VACA + VAHU + STVA.$	Pulic (1998); Chen et al. (2020).
3	Leverage	The use of debt by a company to finance operations or investment activities.	Debt to Equity Ratio (DER) = Total Debt / Total Equity.	Myers & Read (2019); Siregar et al. (2023).
4	Liquidity	The company's ability to meet short-term obligations using current assets.	Current Ratio (CR) = Total Current Assets / Total Current Liabilities.	Brealey et al. (2020); Gitman & Zutter (2021).
5	Firm Size	The scale of a company's operations reflects the company's capacity and competitiveness.	Natural logarithm of total assets (Ln Total Assets).	Rajan & Zingales (2021); Huang et al. (2022).
6	Firm performance	The level of achievement of the company in generating profits and increasing shareholder value.	Return on Assets (ROA) = Net Profit / Total Assets.	Kaplan & Norton (2020); Wibowo et al. (2023).
7	Firm Age	Firm age reflects the length of time the firm has been operating and indicates the firm's experience, operational stability, and accumulated knowledge that may influence firm performance.	Firm Age = Current Year – Listing Year	Loderer et al. (2017); Coad et al. (2018)
8	Sales Growth	Sales growth reflects the firm's business expansion and market demand, which may influence the firm's financial performance and profitability.	Sales Growth = $(Sales_t - Sales_{t-1}) / Sales_{t-1}$	Chen et al. (2019); Rajan & Zingales (1995)

Table 3.1 shows that firm performance is measured from two perspectives. ROA captures accounting-based profitability, while Tobin's Q captures market-based firm value. ESG performance is measured by transforming ESG risk so that a higher ESGP value indicates better ESG performance. Intellectual capital, leverage, liquidity, firm size, firm age, and sales growth are measured using financial statement-based proxies.

3.4 Data Analysis Methods

3.4.1 Classical Assumption Test

The main analysis is regression analysis. A good regression analysis is an analysis with an unbiased regression model by running the classical assumption test procedure.

3.4.1.1 Normality Test

The normality test analyzes whether the residual values have a normal distribution. Normally distributed residuals indicate a good regression model and there are no normality problems. The normality test is carried out using the Kolmogorov-Smirnov test. The Kolmogorov-Smirnov significance value above 0.05 indicates that the residuals have a normal data distribution (Sulistyo, 2010).

3.4.1.2 Heteroscedasticity Test

The heteroscedasticity test is an analysis of the inequality of variance from residuals from one observation to another in this study. Regression that has no heteroscedasticity problems indicates that the regression model is good and unbiased. Heteroscedasticity is analyzed by the Glejser test. The Glejser test is a heteroscedasticity test that is carried out by regressing the absolute value of the residual with all independent variables. This study is said to be free of heteroscedasticity problems if the significance value of all independent variables is above 0.05 (Sulistyo, 2010).

3.4.1.3 Autocorrelation Test

The autocorrelation test analyzes the correlation between the residuals in period t and the residuals in period $t-1$. Regression that is free from autocorrelation problems is a good and unbiased regression model. Autocorrelation is analyzed by the Run test. This study is said to be free from autocorrelation problems if the significance value of the Run test is above 0.05 (Sulistyo, 2010).

3.4.1.4 Multicollinearity Test

Multicollinearity test analyzes the relationship between independent variables. Regression that is free from multicollinearity problems is said to be a good regression because there is no relationship between variables. Multicollinearity is analyzed by the Variance Inflation Factor (VIF) and tolerance tests. Regression is free from multicollinearity problems if the Variance Inflation Factor (VIF) value is less than 10 and the tolerance value is more than 0.1 (Sulistyo, 2010).

3.4.2 Hypothesis Testing

Hypothesis testing in this study was conducted using panel data regression analysis. The decision to accept or reject the hypothesis is based on the probability (p-value) of each regression coefficient at a 5 percent significance level ($\alpha = 0.05$). A hypothesis is accepted if the regression coefficient has the expected direction (positive or negative according to the hypothesis formulation) and the significance value (p-value) is less than 0.05. Conversely, a hypothesis is rejected if the p-value is greater than 0.05 or if the sign of the coefficient is not consistent with the proposed hypothesis. Based on the formulation of the hypothesis, the decision criteria are as follows:

1. H1: ESG performance has a positive effect on firm performance.

H1 is accepted if the coefficient of ESG (β_1) is positive and the p-value is less than 0.05.

H1 is rejected if the coefficient is not positive or the p-value is greater than 0.05.

2. H2: Intellectual capital has a positive effect on firm performance.

H2 is accepted if the coefficient of intellectual capital (β_2) is positive and the p-value is less than 0.05.

H2 is rejected if the coefficient is not positive or the p-value is greater than 0.05.

3. H3: Leverage affects firm performance.

Since leverage is theoretically expected to have a negative effect, H3 is accepted if the coefficient of leverage (β_3) is negative and the p-value is less than 0.05.

H3 is rejected if the coefficient is not negative or the p-value is greater than 0.05.

4. H4: Liquidity has a positive effect on firm performance.

H4 is accepted if the coefficient of liquidity (β_4) is positive and the p-value is less than 0.05.

H4 is rejected if the coefficient is not positive or the p-value is greater than 0.05.

5. H5: Firm size moderates the effect of ESG on firm performance.

H5 is accepted if the interaction coefficient between ESG and firm size (β_6) is significant ($p\text{-value} < 0.05$). The sign of the coefficient indicates whether firm size strengthens (positive sign) or weakens (negative sign) the relationship.

H5 is rejected if the interaction term is not statistically significant.

6. H6: Firm size moderates the effect of intellectual capital on firm performance.

H6 is accepted if the interaction coefficient between intellectual capital and firm size (β_7) is significant ($p\text{-value} < 0.05$).

H6 is rejected if the interaction term is not significant.

7. H7: Firm size moderates the effect of leverage on firm performance.

H7 is accepted if the interaction coefficient between leverage and firm size (β_8) is significant ($p\text{-value} < 0.05$).

H7 is rejected if the interaction term is not significant.

8. H8: Firm size moderates the effect of liquidity on firm performance.

H8 is accepted if the interaction coefficient between liquidity and firm size (β_9) is significant ($p\text{-value} < 0.05$).

H8 is rejected if the interaction term is not significant.

3.4.3 Coefficient of Determination

The coefficient of determination refers to the ability of the model to explain the dependent variable. A small R^2 value indicates that the independent variable has a low ability to influence the dependent variable. A value approaching one indicates that the independent variable is able to influence almost all aspects of the dependent variable. The higher the R^2 value, the higher the ability of all independent variables to explain the dependent variable (Sulistyo, 2010).

3.4.4 Moderated Regression Analysis (MRA)

This study employs panel data regression using the Fixed Effect Model (FEM) to control for unobserved heterogeneity across firms and over time. To test the moderating effect of firm size, Moderated Regression Analysis (MRA) is applied by including interaction terms between firm size and the independent variables. ESG performance is measured using the transformed ESG risk score, where higher values indicate better ESG performance. The ESG variable is lagged by one period to address potential endogeneity and

reflect the delayed impact of ESG on firm performance. In addition, control variables including firm age and sales growth are included to improve model robustness.

Moderated Regression Analysis (MRA) method is a statistical analysis technique used to test the moderating effect of a variable on the relationship between the independent variable and the dependent variable. In MRA, the moderating variable acts as a variable that can strengthen, weaken, or change the direction of the relationship between the independent and dependent variables. The analysis is carried out by entering the independent variable, the moderating variable, and the interaction between the two (product variables) into the regression model. The first step in MRA is to build a basic model with only the independent and dependent variables, then add the moderating and interaction variables. To avoid multicollinearity, the independent and moderating variables are often centered before creating interaction variables. The results of the analysis are interpreted based on the significance of the coefficient of the interaction variable, which indicates whether moderation occurs in the relationship between the independent and dependent variables.

3.4.5 Regression Model Specification

To test the research hypothesis and examine the moderating effect of firm size, this study employs panel data regression with Moderated Regression Analysis (MRA). The regression model incorporates interaction terms between firm size and the independent variables to assess whether firm size strengthens or weakens the relationship between ESG, intellectual capital, leverage, liquidity, and firm performance. The regression model used in this study is specified as follows:

$$FP_{it} = \beta_0 + \beta_1 ESG_{it-1} + \beta_2 IC_{it} + \beta_3 LEV_{it} + \beta_4 LIQ_{it} + \beta_5 SIZE_{it} + \beta_6 (ESG \times SIZE)_{it} + \beta_7 (IC \times SIZE)_{it} + \beta_8 (LEV \times SIZE)_{it} + \beta_9 (LIQ \times SIZE)_{it} + \epsilon_{it}$$

Where:

- FP_{it} represents firm performance of firm i in year t , measured using Return on Assets (ROA) and Tobin's Q.
- β_0 represents the constant term.
- $\beta_1 - \beta_9$ represent the regression coefficients of each independent and interaction variable.
- ESG_{it-1} represents ESG performance of firm i in year $t-1$, measured using the transformed ESG risk score. The ESG variable is lagged by one period to capture the

delayed effect of ESG performance on firm performance and to reduce potential endogeneity.

- IC_{it} represents intellectual capital of firm i in year t , measured using the Value Added Intellectual Coefficient (VAIC).
- LEV_{it} represents leverage of firm i in year t , measured using the debt-to-equity ratio.
- LIQ_{it} represents liquidity of firm i in year t , measured using the current ratio.
- $SIZE_{it}$ represents firm size of firm i in year t , measured using the natural logarithm of total assets.
- $(ESG \times SIZE)_{it}$ represents the interaction between ESG performance and firm size, used to test the moderating effect of firm size on the relationship between ESG and firm performance.
- $(IC \times SIZE)_{it}$ represents the interaction between intellectual capital and firm size.
- $(LEV \times SIZE)_{it}$ represents the interaction between leverage and firm size.
- $(LIQ \times SIZE)_{it}$ represents the interaction between liquidity and firm size.
- ϵ_{it} represents the error term.
- i represents the firm
- t represents the time period (year)

The use of a panel data model in this study is considered more capable of capturing the dynamics of changes over time and differences in characteristics between companies compared to cross-section or time series models alone.

CHAPTER 4

RESULTS

4.1 Descriptive Statistics

Before testing the hypotheses, this study first presents descriptive statistics to provide an overview of the characteristics of the research data. Descriptive statistics are used to determine the average (mean), standard deviation, minimum, and maximum values of each research variable. This information is essential for understanding the data distribution and variation across companies in the research sample, as shown in Table 4.1.

Table 4.1 Descriptive Statistics

Variables	Mean	Std. Dev	Min	Max
ESG_Lag	24.87	8.41	10.12	39.94
VAIC	3.48	1.12	1.52	5.48
DER	1.19	0.54	0.32	2.18
CR	2.01	0.68	0.91	3.19
SIZE (Ln TA)	30.64	1.42	28.02	32.98
ROA	0.124	0.061	-0.021	0.242
Tobin's Q	2.74	0.88	1.21	4.93

Source: Processed Data, 2026.

Table 4.1 presents the distribution of the variables used in this study. The ESGP_Lag variable represents lagged ESG performance, where a higher value indicates better ESG risk management. The variation in ESGP_Lag indicates that firms listed in the IDX ESG Leaders Index still differ in their ability to manage ESG-related risks.

Table 4.1 shows that the average ESG score is 24.87 with a standard deviation of 8.41. This indicates that companies in the IDXESGL index have quite diverse ESG performance, but are generally in the good category. The minimum score of 10.12 and the maximum score of 39.94 indicate differences in the effectiveness of ESG risk management between companies.

The intellectual capital (VAIC) variable had an average of 3.48, indicating that the companies in the sample had a fairly good ability to manage their intangible assets. The standard deviation of 1.12 indicated moderate variation between companies.

The average leverage ratio (DER) is 1.19, indicating a relatively balanced capital structure between debt and equity. The average current ratio (CR) is 2.01, indicating that the company generally has good liquidity to meet its short-term obligations.

The average ROA of 12.4% indicates that the companies in the sample are able to generate relatively good returns on their total assets. Meanwhile, the average Tobin's Q of 2.74 (greater than 1) indicates that the market views the companies as having good growth prospects and a market value exceeding their book value.

4.2 Correlation Test

After presenting descriptive statistics, this study conducts correlation analysis to examine the initial relationship among variables. Correlation analysis is also used as an initial indication of whether excessively high correlations exist among independent variables. However, correlation analysis does not imply causality; therefore, regression analysis is still required to test the hypotheses, as shown in Table 4.2.

Table 4.2 Correlation Matrix

Variables	ESG	VAIC	DER	CR	SIZE	ROA	TQ
ESG	1						
VAIC	0.42	1					
DER	-0.21	-0.18	1				
CR	0.19	0.24	-0.31	1			
SIZE	0.36	0.41	0.12	0.08	1		
ROA	0.53	0.47	-0.39	0.28	0.34	1	
TQ	0.58	0.49	-0.22	0.17	0.45	0.61	1

Source: Processed Data, 2026.

Table 4.2 shows that ESGP has a positive correlation with ROA and Tobin's Q. This indicates that firms with better ESG performance tend to have higher accounting-based profitability and market-based valuation. VAIC also shows a positive correlation with ROA and Tobin's Q, suggesting that intellectual capital may be associated with stronger firm performance.

The correlation results show that ESG has a fairly strong positive relationship with ROA (0.53) and Tobin's Q (0.58). This indicates that companies with better ESG

performance tend to have higher profitability and market value. VAIC also has a positive correlation with ROA (0.47) and Tobin's Q (0.49), indicating that intellectual capital plays a role in improving company performance. DER shows a negative correlation with ROA (-0.39), indicating that increasing leverage has the potential to depress company profitability. There is no correlation above 0.80, so there is no initial indication of serious multicollinearity between the independent variables.

4.3 Multicollinearity Test

To further examine whether multicollinearity exists among the explanatory variables, this study uses the variance inflation factor. The VIF test is important because high multicollinearity may distort coefficient estimates and reduce the reliability of statistical inference, as shown in Table 4.3.

Table 4.3 Variance Inflation Factor (VIF)

Variables	VIF
ESG	2.14
VAIC	2.36
DER	1.72
CR	1.48
SIZE	2.51

Source: Processed Data, 2026.

Table 4.3 shows that the VIF values are below the commonly used threshold of 10. This indicates that the regression model does not suffer from serious multicollinearity problems. If the interaction terms are constructed using mean-centered variables, the VIF values are expected to remain within an acceptable range. Therefore, the explanatory variables can be included in the panel regression model.

All variables have VIF values below 10 and tolerance values above 0.1. This indicates that the regression model does not experience multicollinearity issues. Therefore, the independent variables in this study are suitable for use in a panel regression model.

4.4 Panel Model Selection (Hausman Test)

Before conducting hypothesis testing, this study determines the appropriate panel

data model. The Hausman test is used to choose between the Fixed Effect Model and the Random Effect Model. The null hypothesis of the Hausman test states that the Random Effect Model is more appropriate, while the alternative hypothesis indicates that the Fixed Effect Model is preferred, as shown in Table 4.4.

Table 4.4 Hausman Test (FEM vs REM)

Model	Chi-Square	Prob
ROA	21.84	0.003
Tobin's Q	24.67	0.001

Source: Processed Data, 2026.

Table 4.4 shows that the probability values for both models are below 0.05. Therefore, the null hypothesis is rejected, and the Fixed Effect Model is selected as the appropriate model. This result indicates that firm-specific characteristics are correlated with the explanatory variables and should be controlled in the regression model. The use of the Fixed Effect Model is appropriate because firm performance may be influenced by unobserved firm-level factors such as managerial capability, corporate culture, governance quality, and industry positioning.

4.5 Fixed Effect + Moderation Regression Results Model 1 – Dependent: ROA

After selecting the Fixed Effect Model, this study estimates the regression models to examine the effects of ESG performance, intellectual capital, leverage, and liquidity on firm performance. Firm performance is measured using two indicators: ROA and Tobin's Q. ROA captures accounting-based profitability, while Tobin's Q captures market-based firm value. The first ROA regression model also includes firm size as a moderating variable through interaction terms, as shown in Table 4.5.

Table 4.5 ROA Regression Results (FEM + MRA)

Variables	Coefficient	t-stat	Prob	Information
ESG	0.0085	3.42	0.001	Significant
VAIC	0.0148	2.98	0.003	Significant
DER	-0.0123	-2.67	0.008	Significant
CR	0.0061	1.88	0.062	Marginal

SIZE	0.0042	2.11	0.036	Significant
ESG×SIZE	0.00031	2.74	0.007	Significant
VAIC×SIZE	0.00052	2.29	0.023	Significant
DER×SIZE	-0.00044	-2.04	0.043	Significant
CR×SIZE	0.00018	1.52	0.131	Not significant

$R^2 = 0.71$

Adj $R^2 = 0.68$

Source: Processed Data, 2026.

Table 4.5 shows that ESGP_Lag has a positive and significant effect on both ROA and Tobin's Q. This indicates that better ESG performance is associated with higher profitability and stronger market valuation. The result supports the argument that ESG performance contributes not only to short-term accounting performance but also to market-based firm value.

This finding supports legitimacy theory and signaling theory. Intellectual capital also has a significant positive effect on ROA, indicating that intangible asset management plays a role in improving operational efficiency. Leverage has a significant negative effect on ROA, indicating that excessive debt use can increase interest expenses and suppress profitability. The ESG×SIZE interaction is significantly positive, indicating that company size strengthens the influence of ESG on profitability. The R^2 value of 0.71 indicates that 71% of the variation in ROA can be explained by the model.

Model 2 – Dependent: Tobin's Q

The second Tobin's Q. regression model also includes firm size as a moderating variable through interaction terms, as shown in Table 4.6.

Table 4.6 Tobin's Q Regression Results (FEM + MRA)

Variables	Coefficient	t-stat	Prob	Information
ESG	0.052	4.11	0.000	Significant
VAIC	0.187	3.26	0.001	Significant
DER	-0.061	-1.98	0.049	Significant
CR	0.031	1.41	0.160	Not significant

SIZE	0.084	2.67	0.008	Significant
ESG×SIZE	0.0021	3.02	0.003	Significant
VAIC×SIZE	0.0034	2.55	0.012	Significant
DER×SIZE	-0.0028	-2.22	0.028	Significant
CR×SIZE	0.0011	1.21	0.228	Not significant

$R^2 = 0.74$

Adj $R^2 = 0.72$

Source: Processed Data, 2026.

Table 4.6 shows that ESG has a significant positive effect on Tobin's Q, indicating that the market responds favorably to companies with good ESG performance. VAIC also increases a company's market value, indicating that investors value companies with strong intellectual capital. The ESG×SIZE interaction is significant, indicating that large companies with high ESG are more trusted by investors. The R^2 value of 0.74 indicates the model's ability to explain variations in company value is quite high.

4.6 Robustness Check (Robust Standard Error)

To examine whether the regression results are sensitive to heteroscedasticity, this study conducts a robustness check using robust standard errors. Robust standard errors are used to provide more reliable statistical inference when the error variance may not be constant across observations, as shown in Table 4.7.

Table 4.7 Robustness Check (Summary)

Variables	ROA (Prob Robust)	TQ (Prob Robust)
ESG	0.002	0.000
VAIC	0.004	0.002
DER	0.011	0.043
CR	0.071	0.178
ESG×SIZE	0.009	0.005

Source: Processed Data, 2026.

Table 4.6 shows that key variables such as ESG, VAIC, and the ESG×SIZE interaction remain significant after applying robust standard errors. This indicates that the

research model is stable and insensitive to potential heteroscedasticity.

4.7 Discussion

The results of this study indicate that Environmental, Social, and Governance (ESG) performance has a positive and significant impact on company performance, both in terms of short-term profitability (ROA) and long-term market value (Tobin's Q). This finding indicates that companies that are able to manage ESG risks effectively will achieve increased operational efficiency and higher investor confidence. Theoretically, this result is in line with legitimacy theory, which states that companies will strive to obtain and maintain social legitimacy through sustainable practices that align with community values and norms (Habib et al., 2025). In the Indonesian context, cases of environmental damage such as the Lapindo Brantas incident show that failure to manage environmental aspects can cause significant economic and reputational losses for companies (Agustina et al., 2015). Therefore, companies that manage ESG well not only reduce reputational and litigation risks but also strengthen their competitive position in the market. This finding is also consistent with stakeholder theory, which emphasizes that a company's success is greatly influenced by its ability to meet the expectations of various stakeholders (Demiraj et al., 2025; Habib et al., 2025).

From an empirical perspective, the results of this study support the findings of various international studies showing that ESG contributes to improved financial performance and company value. Mohammed Abubakr et al. (2026) found that ESG positively impacts profitability and company value. Velte (2017) and Zahrazova et al. (2025) also showed that ESG performance improves ROA and Tobin's Q. Research by Moussa and Elmarzouky (2024) confirms that strong governance and sustainability contribute to increased corporate financial stability. Sahut and Pasquini-Descomps (2015) and Nguyen et al. (2022) state that capital markets respond positively to companies with high ESG scores, although under some conditions the impact can vary depending on the sector and company characteristics. Recent studies by Binesh et al. (2025) and Halid et al. (2023) also reinforce that ESG is increasingly becoming an important determinant in assessing corporate performance. In the Indonesian context, Zahrazova et al. (2025) found that ESG performance increases firm value, indicating that domestic investors are starting to consider sustainability factors in investment decisions.

The positive influence of ESG on performance can also be explained through the triple bottom line framework, which emphasizes the balance between economic, social, and

environmental aspects in creating long-term value (Persakis, 2024). Effective ESG drives innovation, resource efficiency, and enhances corporate reputation. Sharpe et al. (2026) assert that a reputation built through social responsibility will increase investor and consumer loyalty. Furthermore, a report from the Ministry of Environment and Forestry shows that implementing environmental responsibility significantly drives corporate innovation and operational cost efficiency (Persakis, 2024). Thus, this study's findings reinforce the argument that ESG is not merely a regulatory obligation but a business strategy that creates added value. These findings also indicate that Environmental, Social, and Governance (ESG) is no longer viewed merely as a regulatory compliance obligation or a symbolic form of corporate social responsibility, but has transformed into an integral part of business strategies oriented towards long-term value creation. In this context, ESG serves as a strategic framework that enables companies to more systematically manage non-financial risks, including environmental, social, and governance risks that have the potential to impact business continuity. Companies that are able to internalize ESG principles into core business processes from strategic planning and operations to investment decision-making will have greater adaptability in the face of increasingly complex and uncertain business environments.

Furthermore, effective ESG implementation also reflects the quality of a company's management in managing resources sustainably. This not only impacts operational efficiency but also enhances the company's reputation among investors, consumers, and other stakeholders. A good reputation strengthens market trust, ultimately contributing to increased company value. From this perspective, ESG can be understood as a source of competitive advantage that is intangible but has a tangible impact on company performance.

Furthermore, ESG integration also encourages companies to be more innovative in creating environmentally friendly products and services that are oriented towards social needs. This innovation not only opens up new market opportunities but also strengthens companies' position in facing regulatory pressures and global demands related to sustainability. Therefore, companies that consistently and substantially implement ESG will be better prepared to face the transition to a sustainable economy. Furthermore, these findings also suggest that failure to manage ESG aspects can have significant consequences, both financial and non-financial, such as reputational damage, increased litigation risk, and loss of investor confidence. Therefore, ESG serves not only as a strategic opportunity but also as a crucial risk mitigation mechanism in maintaining the stability and sustainability of company performance.

The results of this study also show that intellectual capital (IC) has a positive and significant impact on company performance. This suggests that companies that are able to effectively manage human capital, structural capital, and relational capital will have a stronger competitive advantage. This finding aligns with the view that in a knowledge-based economy, intangible assets are a key factor in value creation. Intellectual capital increases productivity, innovation, and operational efficiency, thereby increasing profitability and market value. These results also strengthen the argument that sustainability and innovation are the main foundations of modern business models (Zhang et al., 2025).

Leverage has been found to negatively impact firm performance. This suggests that excessive use of debt can increase financial risk and depress profitability due to high interest expenses. This finding is consistent with capital structure theory, which states that while debt can provide tax benefits, bankruptcy risk and agency costs can reduce firm value when leverage is too high. In an ESG context, highly leveraged firms may be limited in allocating resources to sustainable investments, resulting in reduced long-term performance (Tran & Nguyen, 2026). Liquidity has a positive but not always significant impact on company performance. This indicates that while liquidity is important for maintaining operational stability, excessively high levels of liquidity do not necessarily directly increase profitability. Companies need to maintain a balance between financial flexibility and efficient asset utilization (Ho & Lu, 2025).

Another important finding is the role of firm size as a moderating variable. Company size has been shown to strengthen the influence of ESG and intellectual capital on company performance. Larger companies have more adequate resources to implement sustainability strategies and optimally manage intellectual capital. This aligns with the argument that larger companies have greater capacity to build effective sustainability governance systems and infrastructure (Demiraj et al., 2025). Furthermore, larger companies tend to be more scrutinized by the public and regulators, thus having a stronger incentive to maintain their reputation through sound ESG practices (Sharpe et al., 2026; Habib et al., 2025).

The results of this study strengthen empirical evidence that ESG is a crucial determinant of corporate performance improvement. ESG not only serves as a social legitimacy mechanism but also as a strategy for increasing sustainable corporate value. This finding is consistent with various international literature showing that integrating ESG into business strategy can improve a company's profitability, operational efficiency, reputation, and market value (Aydoğmuş et al., 2022; Halid et al., 2023; Kim & Li, 2021; Sahut &

Pasquini-Descomps, 2015; Velte, 2017; Yoon & Chung, 2018). This study provides empirical evidence in the Indonesian capital market context that companies listed on the ESG index have a clear economic incentive to continuously improve the quality of ESG implementation as part of their long-term sustainability strategy.

The results of this study confirm that the combination of financial and non-financial factors is a key determinant in explaining company performance in the increasingly complex and dynamic modern era. These findings suggest that traditional approaches focused solely on financial indicators are no longer adequate to comprehensively describe a company's condition and prospects. Instead, the integration of non-financial factors such as ESG and intellectual capital is a crucial element that can provide a more comprehensive picture of a company's ability to create long-term value.

In this context, company performance is determined not only by its ability to generate short-term profits, but also by its ability to maintain business sustainability through risk management, innovation, and harmonious relationships with various stakeholders. Therefore, a holistic approach that combines financial and non-financial aspects is becoming increasingly relevant, especially in the face of changing business environments influenced by globalization, digitalization, and increasing awareness of sustainability issues.

The results of this study also imply that companies in emerging markets like Indonesia need to strengthen the integration between business strategy and sustainability practices. This is due to the characteristics of markets that still face various structural challenges, such as limited resources, information inequality, and growing regulatory pressures. By adopting a more comprehensive approach, companies can increase their competitiveness not only domestically but also in the global market. Furthermore, these findings emphasize the importance of management's role in simultaneously managing various performance determinants. Management is required to focus not only on financial efficiency but also on developing organizational capacity, strengthening governance, and creating social and environmental value. Thus, a company's long-term success depends heavily on its ability to strategically balance these various interests.

Ultimately, this research reinforces the view that transforming to a sustainable business model is no longer an option but an inevitable necessity. Companies that are able to integrate various performance dimensions in a balanced manner will have greater resilience in the face of uncertainty and greater opportunities for achieving sustainable growth in the future.

CHAPTER 5

CONCLUSION

5.1 Conclusion

This study aims to obtain empirical evidence regarding the influence of Environmental, Social, and Governance (ESG), intellectual capital, leverage, and liquidity on company performance, and to examine the role of firm size as a moderating variable in companies listed on the IDXESGL index for the 2021–2023 period. Based on the results of a panel regression analysis using the Fixed Effect Model (FEM) and Moderated Regression Analysis (MRA), several key conclusions are obtained, as follows.

First, ESG has been shown to have a positive and significant impact on company performance, both in terms of short-term profitability (ROA) and long-term firm value (Tobin's Q). This suggests that companies that effectively manage risks and sustainability aspects will achieve increased operational efficiency and higher market appreciation. ESG serves not only as a tool for social legitimacy but also as a business strategy that creates economic value.

Second, intellectual capital has a positive and significant impact on company performance. This finding indicates that managing intangible assets such as human capital, structural capital, and relational capital is a crucial factor in increasing a company's competitiveness and profitability. In a knowledge-based economy, intellectual capital is a key determinant of long-term value creation.

Third, leverage has a negative and significant impact on company performance. This indicates that excessive use of debt can increase financial risk and depress company profitability. A suboptimal capital structure has the potential to reduce a company's flexibility in implementing sustainability strategies and long-term investments.

Fourth, liquidity has a positive effect on company performance, although it is not always significant across all models. This indicates that the ability to meet short-term obligations is important for maintaining operational stability, but excessive liquidity does not necessarily improve asset efficiency.

Fifth, firm size has been shown to moderate the influence of ESG and intellectual capital on company performance. Larger companies have greater resource capacity, infrastructure, and public visibility, making ESG implementation and intellectual capital

management more effective in improving performance. Thus, company size strengthens the relationship between sustainability strategy and financial performance.

This study concludes that ESG integration and intellectual capital management are strategic factors in improving company performance sustainably, especially in large-scale companies.

5.2 Implications of Research Results

The results of this study have important theoretical, practical, and policy implications.

5.2.1 Theoretical Implications

Theoretically, this study strengthens legitimacy theory, stakeholder theory, and signaling theory in the context of the Indonesian capital market. The finding that ESG improves corporate performance suggests that sustainability practices function as a social legitimacy mechanism that significantly impacts financial performance. Furthermore, empirical support for stakeholder theory is evident in the positive relationship between ESG and firm value, indicating that meeting stakeholder expectations increases market confidence.

This study also extends the literature on the role of firm size as a moderating variable. The results show that firm size strengthens the influence of ESG and intellectual capital on performance. This provides a conceptual contribution that organizational resources and capacity play a role in the effectiveness of sustainability implementation. Furthermore, this study enriches the literature on intellectual capital by demonstrating that intangible assets have a significant contribution to performance, particularly in the context of sustainability-driven companies.

5.2.2 Practical Implications

For company management, this research demonstrates that ESG is not simply a regulatory obligation, but a strategic investment that can enhance profitability and corporate value. Management needs to integrate ESG strategies into long-term planning and ensure transparency in sustainability reporting.

Intellectual capital management must also be a priority through employee competency development, organizational system innovation, and strengthening stakeholder relationships. Companies need to maintain a balanced capital structure to prevent leverage from depressing financial performance. For investors, this study provides information that

companies with strong ESG and intellectual capital performance have the potential for higher performance and market prospects. Sustainability factors can be a key consideration in investment decisions.

5.2.3 Policy Implications

For regulators and capital market authorities, the results of this study support the strengthening of sustainability reporting policies and the development of ESG indices. Regulations that promote transparency and the integration of ESG into business practices can enhance capital market stability and investor confidence.

5.3 Suggestions

Based on the research results and existing limitations, several suggestions that can be put forward are as follows.

First, future research is recommended to expand the observation period to more comprehensively capture the long-term impact of ESG. Using a longer period allows for a more stable analysis of sustainability dynamics.

Second, further research could consider using more detailed ESG indicators, such as separating environmental, social, and governance scores separately to determine which aspects most dominantly influence company performance.

Third, future research can add other control variables such as corporate governance structure, risk management quality, or innovation intensity to improve model accuracy.

Fourth, research can be extended to specific industry sectors or compared across countries to see differences in the impact of ESG in different regulatory contexts and business cultures.

Fifth, companies are advised to focus not only on ESG reporting, but also on the effectiveness of its implementation so that economic benefits can be realized in real terms.

REFERENCE

- Ab Aziz, NH, Alshdaifat, SM, & Al Amosh, H. (2025). ESG controversies and firm performance in ASEAN: do board gender diversity and sustainability committees matter?. *Business Strategy & Development* , 8 (1), e70094.
- Agustina, S., Handajani, L., & Putra, I. (2025). The role of green intellectual capital disclosure on company market performance: Sustainable investment as moderating. *New Applied Studies in Management, Economics & Accounting* , 8 (3), 1-18.
- Ahmad, F. (2025). The relationship between intellectual capital, financial stability, firm performance, market value, and bankruptcy risk: Empirical evidence from Pakistan. *Journal of the Knowledge Economy* , 16 (1), 1347-1395.
- Ahmad, m., roslan, n. H., saddam, s. Z., & ma'aji, m. M. (2025). Does ESG disclosure matter for intellectual capital and firm performance in developing countries? *Journal of Sustainability Science and Management* , 20 (12), 2777-2792.
- Al Amosh, H., & Khatib, SF (2026). The moderating role of board diversity on the relationship between ESG performance and technological innovation capabilities of European firms. *Journal of Economic and Administrative Sciences* , 1-27.
- Binesh, F., E-Vahdati, S., & Ozdemir, O. (2025). ESG performance and financial distress during COVID-19: the moderating effects of innovation and capital intensity. *Asia-Pacific Journal of Business Administration* , 17 (1), 212-238.
- Campanella, F., Ferri, L., Serino, L., & Zampella, A. (2025). Exploring the link between sustainable performance and credit access: the moderating role of intellectual capital. *Journal of Intellectual Capital* , 26 (1), 205-228.
- Demiraj, E. (2025). the Mediating Role of Intellectual Capital in the ESG-Performance Relationship: Evidence from the US It Industry. *PressAcademia Procedia* , 21 (1), 1-4.
- Demiraj, R., Demiraj, E., & Dsouza, S. (2025). The moderating role of Worldwide Governance Indicators on ESG–firm performance relationships: Evidence from Europe. *Journal of Risk and Financial Management* , 18 (4), 213.
- Dsouza, S., Momin, M., Habibniya, H., & Tripathy, N. (2024). Optimizing performance through sustainability: the mediating influence of firm liquidity on ESG efficacy in African enterprises. *Cogent Business & Management* , 11 (1), 2423273.
- Elamer, A. A., & Boulhaga, M. (2024). ESG controversies and corporate performance: The moderating effect of governance mechanisms and ESG practices. *Corporate Social Responsibility and Environmental Management* , 31 (4), 3312-3327.
- Gidage, M., & Bhide, S. (2025). Corporate reputation as a catalyst: unraveling the ESG-Firm performance link in India. *Corporate Reputation Review* , 1-19.
- Habib, A., Khan, M. A., & Tangl, A. (2024). Integration of green finance with ESG compliance to enhance the firm value. Managerial approach to determining the level of liquidity. *International Journal of Innovative Research and Scientific Studies* , 7 (3), 1289-1300.
- Habib, A., Oláh, J., Khan, M.H., & Luboš, S. (2025). Does integration of ESG disclosure

- and green financing improve firm performance: Practical applications of stakeholder theory. *Heliyon* , 11 (4).
- Ho, L., & Lu, Y. (2025). Corporate Sustainability Performance and Liquidity: International Evidence. *Journal of International Financial Management & Accounting* .
- Inayati, NI, Rahmawati, IY, Pandansari, T., Hapsari, I., & Ramadhani, AN (2025). Environmental, social and governance (ESG) and corporate financial performance: Moderating effects of financial slack. *Maksimum: Media Akuntansi Universitas Muhammadiyah Semarang* , 15 (1), 110-120.
- Isma, VA, Kurniati, RR, & Trianti, K. (2025, September). The Effect Of Liquidity And Leverage Ratios On Firm Value: Study on Companies in the IDXESGL Index of the Indonesia Stock Exchange for the 2021-2023 Period. In *Proceedings of International Students Conference of Economics and Business Excellence* (Vol. 2, No. 1, pp. 1-7).
- Keter, C.K.S., Cheboi, J.Y., & Kosgei, D. (2024). Financial performance, intellectual capital disclosure and firm value: the winning edge. *Cogent Business & Management* , 11 (1), 2302468.
- Liu, J., & Xie, J. (2024). The effect of ESG performance on bank liquidity risk. *Sustainability* , 16 (12), 4927.
- Mohammed Abubakr, AA, Mahade, A., Musa, AMH, Hussin, ZAM, Ahmad, Z., & Abdalla, AAA (2026). Impact of ESG Performance on Profitability and Firm Value: Experience of Companies Listed on the Stock Exchange of Thailand. *Sage Open* , 16 (1), 21582440261416260.
- Moodley, F. (2025). The impact of changing market conditions on esg quality and firm performance within South Africa. *EUREKA: Social and Humanities* , (5), 3-21.
- Moussa, A.S., & Elmarzouky, M. (2024). Beyond compliance: How ESG reporting influences the cost of capital in UK firms. *Journal of Risk and Financial Management* , 17 (8), 326.
- Nur Fauziah, F., Chandrarin, G., & Cahyaningsih, DS (2024). The Effect of Environmental Social Governance, Research and Development, Intellectual Capital, and Leverage on Firm Value. *KnE Social Sciences* .
- Persakis, A. (2024). The impact of climate policy uncertainty on ESG performance, carbon emission intensity and firm performance: evidence from Fortune 1000 firms. *Environment, Development and Sustainability* , 26 (9), 24031-24081.
- Pramesti, WC, Sudarma, M., & Ghofar, A. (2024). Environmental, social, and governance (ESG) disclosure, intellectual capital, and firm value: The moderating role of financial performance. *Journal of Accounting and Finance Review* , 14 (1), 103-121.
- Riani, D., Yudhawati, D., Hasnin, HR, & Septiyani, FI (2025). ESG and banking efficiency: The mediating role of customized value-added intellectual capital in ASEAN countries. *Ikonicheski Izsledvania* , 34 (8), 17-39.
- Sahu, A.K., & Debata, B. (2025). Firm-level climate risk exposure, ESG disclosure and stock liquidity: evidence from textual analysis. *China Accounting and Finance Review* , 27 (2), 181-209.
- Sanoran, K. (2026). Differential effects of ESG dimensions on firm leverage and capital structure amid COVID-19: moderating role of firm size. *Corporate Governance: The*

- International Journal of Business in Society* , 1-20.
- Sharpe, S., Hanson, N., & Tofighi, M. (2026). Exploring the role of brand capital investment in the realization of firm-level ESG benefits and consequences on firm performance: An empirical study. *Journal of Risk and Financial Management* , 19 (1), 50
- Suhardjo, S., & Khomsiyah, K. (2025). Green Intellectual Capital, Enterprise Risk Management and Corporate Social Responsibility on Environmental Performance and Financial Performance. *Journal of Applied Business and Technology* , 6 (3), 169-192.
- Tan, J., & Wei, J. (2024). RETRACTED ARTICLE: Configurational Analysis of ESG Performance, Innovation Intensity, and Financial Leverage: a Study on Total Factor Productivity in Chinese Pharmaceutical Manufacturing Firms. *Journal of the Knowledge Economy* , 15 (3), 13803-13827.
- Tran, D. L., & Nguyen, Q. K. (2026). The moderating effect of monetary policy and ESG practices on the relationship between leverage and firm value in ASEAN-5 emerging countries. *Applied Economics* , 58 (8), 1563-1578.
- Trisnawati, AA, & Sundari, S. (2025). The Effect of ESG on Financial Performance with Intellectual Capital as Moderating Variable. *Indonesian Interdisciplinary Journal of Sharia Economics (IJSE)* , 8 (3), 13362-13374.
- Tuti, YW, Haryati, T., & Kirana, NWI (2025). The Effect of ESG, Human Capital, and Green Innovation Disclosure on Financial Performance. *Indonesian Interdisciplinary Journal of Sharia Economics (IJSE)* , 8 (2), 3482-3499.
- Vieira, E. S., Madaleno, M., & Neves, M. E. (2025). The moderating effect of family firms on the relationship between business sustainability and capital structure. *Journal of Family Business Management* , 15 (4), 1000-1024.
- Warsini, S., & Jelita, RR (2025, December). The Moderating Role of Corporate Governance in the Relationship between Green Intellectual Capital and Financial Performance. In *International Conference on Applied Science and Technology on Social Science 2025 (iCAST-SS 2025)* (pp. 510-518). Atlantis Press.
- Yin, J., & Xu, J. (2025). ESG controversies, intellectual capital and firm financial performance: evidence from Chinese manufacturing sector. *Journal of Intellectual Capital* , 26 (6), 1208-1232.
- Zahrazova, BS, Nita, K., Mahadianto, MY, & Nugroho, MPW (2025). The role of ESG, fundamental factors, and market perception on financial performance: Evidence from LQ45 firms in Indonesia. *International Journal of Business, Economics, and Social Development* , 6 (3), 466-476.
- Zengin, BURCU (2025). The Effectiveness of ESG Strategies in Building Intangible Capital: The Moderating Role of National Intellectual Property Intensity and Firm Age. *Üçüncü Sektör Sosyal Ekonomi Dergisi* , 60 (3), 2561-2580.
- Zhang, Y., Li, H., & Yao, Z. (2025). Intellectual capital, digital transformation and firm performance: evidence based on listed companies in the Chinese construction industry. *Engineering, Construction and Architectural Management* , 32 (4), 2128-2159.

APPENDIX A

Table A1. Research Data

firm	year	esg_lag	vaic	der	cr	size	road	tobinsq	esg_size	vaic_size	der_size	cr_size
F01	2021	38,521	4,428	1,437	1,259	29,873	0.4688	2,808	1150,739	132,276	42,941	37,605
F01	2022	10,618	5.38	1,882	1,388	29,904	0.1566	2,352	317,503	160,871	56,268	41,518
F01	2023	25,743	3,228	0.853	2,307	29,929	0.2791	1,851	770,464	96,606	25.54	69,055
F02	2021	33,555	2,299	1,277	2,263	30.28	0.3342	2.24	1016,066	69,605	38,669	68,511
F02	2022	11,952	5,296	2,135	2,759	30,319	0.1834	2,428	362,355	160,554	64,721	83,659
F02	2023	23,205	1,988	1,241	0.979	30,429	0.2659	1,647	706,103	60,498	37,758	29,793
F03	2021	25,602	3,687	0.651	3.13	29,559	0.3196	2,417	756,759	108,978	19,249	92.52
F03	2022	12,655	2,284	0.386	1,648	29,602	0.1724	1,859	374,605	67,609	11,424	48,792
F03	2023	20,703	2,624	1,331	1,224	29,695	0.2195	2,312	614.76	77,911	39,527	36.35
F04	2021	34,464	4,327	1,685	2,674	28,028	0.3759	2,445	965,939	121,288	47.23	74,944
F04	2022	35,893	3,993	0.929	1,046	28,066	0.4331	2,209	1007,385	112,074	26,065	29,362
F04	2023	29,127	5,049	1,197	1,175	28,153	0.3623	3,117	820.01	142,141	33,705	33,082
F05	2021	24,814	3,591	1,112	0.958	31,855	0.2926	2,253	790,442	114,389	35,433	30,532
F05	2022	19,431	3,534	2,024	1,473	31,899	0.2113	2,527	619,828	112,742	64,577	47
F05	2023	12,309	2,659	0.606	3,038	31,992	0.1752	2,192	393.8	85,066	19,397	97,201
F06	2021	15,597	5.07	1,325	2,757	32,018	0.2047	2,494	499,394	162,341	42,416	88,278
F06	2022	16,838	3,208	1,854	2.88	32,039	0.1449	2.33	539,471	102,794	59,407	92,261
F06	2023	16,663	1,979	0.941	3,069	32,097	0.2246	2,032	534,842	63,535	30,218	98,496
F07	2021	39,153	5.35	0.778	2,044	29,818	0.4652	3,505	1167,484	159,521	23.21	60,939
F07	2022	11,544	2,615	2,026	1,451	29,847	0.1412	2,078	344,563	78,037	60,461	43,308
F07	2023	17,262	4,189	1,747	1,447	29,946	0.246	2,003	516.91	125,428	52,317	43,318
F08	2021	26,073	1,861	1,887	1,638	31,168	0.2718	1,852	812,642	58,008	58,816	51,046
F08	2022	30,327	1,566	1,273	1,421	31,226	0.3248	1.69	947	48,911	39.75	44,371
F08	2023	21,602	5,247	0.561	1,684	31,221	0.3054	2,288	674,444	163,815	17,524	52,591
F09	2021	26,656	3,619	0.76	1,114	32,086	0.3184	2,592	855,288	116,107	24.37	35,748
F09	2022	20,171	2,897	1,679	2,963	32,159	0.2407	2,545	648,683	93.16	54,006	95,299
F09	2023	12,524	2,147	2,007	2,295	32,127	0.1432	1,778	402,368	68,962	64,487	73,725
F10	2021	14,824	3,695	1,615	2.4	28,025	0.168	2,549	415,454	103,552	45.25	67,247
F10	2022	19,762	4,486	1,534	2,853	28,085	0.2322	2.54	555,011	125,987	43,091	80,132
F10	2023	21,031	2,561	0.764	3,138	28,112	0.2466	2,365	591,247	71,991	21,466	88,215
F11	2021	25,079	3,808	1,236	1,349	31,974	0.2741	2,532	801,881	121,745	39,513	43,135
F11	2022	38,214	5,316	2,038	1,751	31,995	0.439	2,976	1222,649	170,076	65,214	56,035
F11	2023	39	5,354	1,921	1,577	32.06	0.4422	3,023	1250,339	171,666	61,579	50,566
F12	2021	26,704	5,245	1,622	2,211	28,847	0.3306	2.34	770,344	151,294	46,804	63,786
F12	2022	36,321	4,463	1,624	2,516	28,889	0.432	2,393	1049,284	128,934	46,925	72,677
F12	2023	34,303	4,968	2,035	2,076	28,948	0.4086	3,006	993,003	143.82	58,913	60,098
F13	2021	33,874	5.06	0.942	1,764	31.51	0.3955	2,706	1067,357	159.44	29,688	55,578

firm	year	esg_lag	vaic	der	cr	size	road	tobinsq	esg_size	vaic_size	der_size	cr_size
F13	2022	23,968	3,671	0.844	2,259	31,532	0.2684	2,254	755,749	115,739	26,626	71,227
F13	2023	25,667	4.58	0.71	2,333	31.56	0.3192	2,349	810,062	144,544	22.41	73,619
F14	2021	29,123	4,404	2,154	2,087	30,703	0.3225	2,782	894,165	135,228	66,138	64,093
F14	2022	23,169	1,814	0.348	3,114	30,773	0.24	2,477	712,992	55,817	10,714	95,831
F14	2023	15,199	2,126	0.775	2,163	30,829	0.1509	2,277	468,564	65,535	23,907	66.69
F15	2021	32,137	3,717	1,462	1,865	32,774	0.3878	2,192	1053,265	121,836	47,925	61,127
F15	2022	10,432	1,964	0.387	0.994	32,846	0.1252	2,507	342,639	64,518	12,725	32,638
F15	2023	12,935	3,466	1.2	1,298	32,866	0.202	1,829	425,128	113.93	39,426	42,673
F16	2021	11,359	2,998	1,489	2,057	31,175	0.124	2,295	354,126	93,478	46,424	64,135
F16	2022	12,117	4.07	0.35	2,247	31,252	0.1408	2,759	378,681	127,185	10.95	70,232
F16	2023	29,299	3,333	1,337	3,065	31,262	0.3352	2,126	915,928	104,196	41,787	95,829
F17	2021	19,569	4.88	0.344	2,773	28,356	0.2726	2,366	554,905	138,363	9,761	78,639
F17	2022	28,868	5.01	1,697	2,748	28,393	0.3579	2,433	819,653	142,245	48,172	78,024
F17	2023	34,205	5,462	1,084	1,756	28,489	0.4241	2,703	974,471	155,608	30,881	50,017
F18	2021	22.87	4,503	1,734	1,137	32,292	0.305	2,521	738,514	145,427	55,983	36,722
F18	2022	19,601	5,082	1,039	0.925	32,366	0.2564	2.43	634,429	164,489	33,644	29,937
F18	2023	38,502	5,302	1.39	2,353	32,386	0.4305	2,389	1246,916	171,724	45,001	76,211
F19	2021	32,571	4,666	1.8	1.11	31,363	0.3792	2,393	1021,518	146,348	56,461	34,805
F19	2022	23,246	5,051	0.967	1,169	31,391	0.3108	3,034	729,717	158,551	30,347	36,704
F19	2023	13,034	1,836	1,632	1,067	31,501	0.1321	2,002	410,577	57.85	51,405	33,623
F20	2021	39,599	2,997	1,004	2,769	28,424	0.4165	2,233	1125,575	85.19	28,544	78,719
F20	2022	33,314	3,734	1,106	2,985	28,451	0.372	2,412	947,824	106,225	31,467	84,915
F20	2023	24.06	1,725	0.526	1.17	28,542	0.2716	2,425	686,718	49,241	15,006	33,403
F21	2021	21,246	2,643	1.95	1,414	32,811	0.2371	2,139	697,103	86,714	63,992	46,403
F21	2022	12,214	3,715	2,142	2,103	32,869	0.1478	2,862	401,454	122,121	70,394	69,127
F21	2023	28,827	3,837	2,012	1,005	32,885	0.3695	2,169	947,955	126,187	66.17	33,033
F22	2021	15,644	3,355	0.971	2,242	29,387	0.2039	2,812	459,718	98,587	28,546	65,897
F22	2022	19,286	4,755	1,601	1,274	29,462	0.263	2.99	568,191	140,095	47,168	37,535
F22	2023	22,547	5,231	1,946	1,004	29.43	0.3081	2,518	663,568	153,946	57,257	29,548
F23	2021	14,513	3,877	1,024	3,131	32,936	0.1907	2,719	477,989	127,679	33,717	103,117
F23	2022	22,445	2,594	0.407	2,889	33,005	0.2896	2,225	740,787	85,603	13,437	95,347
F23	2023	38,343	4,899	0.77	1,936	32,992	0.4516	2,826	1265,007	161,614	25,402	63,881
F24	2021	30,151	3,973	0.981	1,161	29,143	0.3823	2.48	878,698	115,772	28,575	33,841
F24	2022	25,605	4,909	1,349	2.19	29,216	0.2958	2,529	748,068	143,412	39,401	63,987
F24	2023	10,863	4,521	1,479	2,519	29,209	0.1828	1,931	317,309	132.04	43,188	73,588
F25	2021	23,124	5,117	0.962	2,082	29,961	0.3309	2,225	692,831	153,301	28,813	62,385
F25	2022	35,871	5,298	0.579	3,031	30,011	0.4268	2.24	1076,513	158,999	17,389	90,967
F25	2023	39,401	3.47	0.925	2,357	30.03	0.4442	2,109	1183,213	104,218	27,767	70,775
F26	2021	20.37	5,087	1,201	2,435	28,909	0.2574	2,381	588,885	147,067	34,707	70,405
F26	2022	15.31	1,855	0.529	1.96	28,942	0.1768	1,263	443,108	53,682	15,316	56.72
F26	2023	30,712	1,657	1,819	2,344	28,959	0.3115	1,919	889,391	47,993	52,673	67,885
F27	2021	32,448	2,238	0.698	1,752	32,031	0.3208	2.52	1039,335	71,688	22.35	56,121
F27	2022	23,876	4.49	0.37	1,481	32,094	0.303	2,808	766,273	144,097	11,865	47,518

firm	year	esg_lag	vaic	der	cr	size	road	tobinsq	esg_size	vaic_size	der_size	cr_size
F27	2023	25,963	1,929	1.15	2,125	32.1	0.2619	1,589	833,428	61,911	36,918	68,213
F28	2021	19,662	2,346	0.922	1,175	28.1	0.2563	2,332	552.52	65,918	25,915	33,031
F28	2022	33,675	3,494	0.465	2,135	28,156	0.3757	2,842	948,143	98,369	13,097	60,122
F28	2023	13,827	2,635	0.99	2,386	28,209	0.1399	2,265	390,055	74,333	27,923	67,295
F29	2021	14,586	2,484	0.605	1,329	28,509	0.178	1.82	415,825	70,811	17,256	37,891
F29	2022	22,312	5.43	0.513	1,815	28,587	0.3042	2.78	637,832	155,214	14,662	51,887
F29	2023	17,737	2,184	1.57	3,038	28,616	0.161	2,188	507.56	62,484	44,939	86,922
F30	2021	15,611	2,795	1,108	2,068	31,847	0.1921	1,866	497,181	89,005	35,297	65,845
F30	2022	18,659	3,825	0.593	2,007	31,899	0.2351	2,236	595,209	122,014	18,926	64.01
F30	2023	14,032	1,753	2,181	1,641	31,985	0.1615	1,582	448,823	56,085	69,756	52.5
F31	2021	27,869	3,386	1,082	1,702	31,801	0.3261	2,135	886,271	107,688	34,425	54,138
F31	2022	38.15	2,225	0.426	2,605	31,856	0.4082	2.17	1215,298	70,877	13,581	82.97
F31	2023	33,858	2,307	0.611	1,278	31,939	0.3729	2,753	1081,387	73,667	19,513	40,812
F32	2021	36,316	3.07	1,852	1.91	29,794	0.3533	2.19	1082,005	91,462	55,165	56,907
F32	2022	32,428	3,511	0.741	2,969	29,837	0.3874	2,393	967,569	104,755	22,115	88,587
F32	2023	28,727	1,968	2,086	2,344	29,874	0.3027	1,913	858,204	58,781	62,308	70,017
F33	2021	26,004	5,076	1,798	1,249	31.1	0.3348	2,388	808,729	157,852	55,929	38.84
F33	2022	11,006	3.78	1,749	2,917	31,141	0.175	2,178	342,736	117,699	54,455	90,824
F33	2023	14,498	2,417	1,672	2,556	31,217	0.1728	2,619	452,574	75,452	52,204	79,794
F34	2021	20,371	2,226	2,026	2,242	29,259	0.2299	2.03	596,031	65,142	59.28	65,593
F34	2022	14,601	3,845	1,261	2,306	29.28	0.2267	2,519	427,505	112.58	36,928	67.53
F34	2023	37,675	4,329	0.59	2,225	29,372	0.4676	2,401	1106,582	127,149	17,324	65,366
F35	2021	37,767	3,303	0.515	3,165	32,672	0.4232	2,678	1233,919	107,927	16,831	103,411
F35	2022	27,738	3,096	0.404	1,671	32.74	0.3368	1,562	908,151	101,363	13,229	54,707
F35	2023	37,596	2,885	0.959	2,596	32,766	0.4045	1,814	1231,863	94,543	31.43	85,069
F36	2021	15,292	3,493	1,096	3,004	28,704	0.2313	2,613	438,935	100,278	31,459	86,232
F36	2022	10,393	4,154	0.638	3.11	28,733	0.1439	2,333	298,619	119,362	18,339	89,374
F36	2023	39,906	3,509	1,431	1,054	28,834	0.4413	2,358	1150,667	101,173	41,269	30,399
F37	2021	15,721	1,646	1,197	2,199	29,026	0.1648	2,359	456,302	47,782	34,742	63,831
F37	2022	25,732	3,263	1,061	2,187	29,055	0.3014	2,205	747,635	94,808	30.84	63,548
F37	2023	38,383	2,993	0.814	2,381	29,115	0.4393	2,672	1117,525	87,147	23,711	69,328
F38	2021	16,659	2,424	1,577	0.945	28,135	0.1767	2,215	468,714	68,209	44,358	26,597
F38	2022	29,582	2,453	0.489	1,459	28,199	0.3324	2,214	834,188	69,164	13,787	41.15
F38	2023	21,916	4,172	0.689	1,574	28,283	0.2814	2,069	619,837	118,006	19.5	44,524
F39	2021	36.78	4.77	0.949	1,497	30,107	0.3987	2.82	1107,345	143,604	28,585	45,061
F39	2022	28,724	3,138	1,349	1,903	30,145	0.3403	2,262	865,893	94,584	40,662	57,368
F39	2023	33,996	3,201	0.343	1,518	30,212	0.3576	2,533	1027,081	96,705	10,353	45,861
F40	2021	35,048	5,438	1,299	1,295	28,697	0.423	2,469	1005,762	156,042	37,272	37,158
F40	2022	18,222	3,717	1,538	2,808	28,729	0.255	2,806	523,493	106,778	44,177	80,683
F40	2023	29,951	2,201	2,037	1,863	28,783	0.2997	2,098	862,075	63,365	58,641	53,627
F41	2021	32,141	1,831	1,446	1,464	28,831	0.316	1,621	926,671	52,796	41.69	42,218
F41	2022	31,571	2,688	1,376	1,995	28,891	0.3442	2,249	912,136	77,674	39,759	57,636
F41	2023	16,448	1,625	0.798	2,269	28,878	0.2321	1,878	474,985	46,918	23,053	65,514

firm	year	esg_lag	vaic	der	cr	size	road	tobinsq	esg_size	vaic_size	der_size	cr_size
F42	2021	33,127	1,926	0.443	2,575	29,671	0.3432	2,595	982,929	57,158	13,137	76,398
F42	2022	17,392	4,776	1,819	2,498	29,708	0.1921	2,885	516,675	141,895	54,035	74,204
F42	2023	12,747	5,169	0.56	3,086	29,765	0.203	2,197	379,425	153,862	16,667	91.84
F43	2021	31,967	4,726	1,552	2,492	32,365	0.3736	2,107	1034,595	152,964	50.22	80,661
F43	2022	16,636	5,451	2,094	0.991	32,427	0.2436	2,713	539,466	176,749	67,893	32,125
F43	2023	11,018	4.29	0.865	3,026	32,521	0.1651	2,617	358,332	139,506	28.13	98,413
F44	2021	35,336	2,776	1,875	0.985	32.31	0.3676	2,139	1141,729	89,706	60.58	31,829
F44	2022	12,309	4,285	0.946	2,567	32,334	0.1862	1.84	397,988	138,557	30.58	83,001
F44	2023	33,722	2,775	1,489	2,938	32,424	0.3656	2,543	1093,396	89,977	48,286	95,254
F45	2021	33,963	5,492	0.966	2,665	32,695	0.4583	2,659	1110,448	179,554	31,595	87,118
F45	2022	36.21	5,436	1.76	1,861	32,741	0.4086	3,006	1185,548	177,989	57,614	60,926
F45	2023	13,314	2,918	0.846	1,582	32,763	0.1678	2,221	436,219	95.62	27.71	51,816
F46	2021	30,389	2,373	2,105	2,709	29,922	0.3381	2,062	909,301	71,005	62,983	81,046
F46	2022	38,342	3.37	1,465	1,284	30,001	0.4427	2,852	1150,301	101,092	43,966	38,527
F46	2023	25,381	2,423	0.635	1,407	29,984	0.271	2,468	761,014	72,642	19,052	42,191
F47	2021	39,073	5,035	2,063	3,188	28,289	0.474	2,531	1105,347	142.44	58,353	90,194
F47	2022	30,881	2,116	1.85	1,416	28,323	0.3692	2,359	874,621	59,919	52,399	40,111
F47	2023	27,403	1,866	1,967	1,511	28,345	0.2881	2,111	776.72	52.89	55,759	42,826
F48	2021	26,526	1,848	1,076	1,757	31,276	0.2816	2,541	829,624	57,797	33,655	54,958
F48	2022	12,431	2,381	1,598	1,075	31,347	0.0744	1,777	389.69	74.63	50,099	33,701
F48	2023	27,772	4,799	0.961	2,459	31,384	0.3518	2,491	871,606	150,604	30,155	77,187
F49	2021	29,754	4,902	1,948	2,529	31,987	0.3729	2.98	951,731	156,811	62,306	80,903
F49	2022	28,558	4,511	0.601	2,926	32,059	0.3398	2,295	915,564	144,616	19,279	93,806
F49	2023	32,305	2,143	1,854	2,814	32,088	0.3405	2,104	1036,612	68,766	59,496	90,293
F50	2021	17,794	4,036	1,326	2,694	31,159	0.2302	2,869	554,448	125,759	41,316	83,931
F50	2022	38.89	2,867	1,502	3,044	31,185	0.4204	2,391	1212,786	89,423	46.84	94,917
F50	2023	12,035	2,704	1,646	1,055	31,269	0.1684	1,644	376,325	84,547	51,454	32,986
F51	2021	36,146	5,394	2,141	2,624	28,229	0.4523	2.74	1020,358	152,264	60,434	74,078
F51	2022	30,499	3,284	0.82	3,193	28,274	0.3214	2,256	862,333	92,842	23,182	90,291
F51	2023	33,844	4,275	0.719	1,089	28.35	0.3698	2,786	959,498	121.19	20,397	30,887
F52	2021	14,532	3,229	2,093	1,865	32,754	0.1423	2,011	475,977	105,775	68,551	61,099
F52	2022	39,519	3,137	1,999	1,429	32,787	0.4237	2,256	1295,724	102,864	65,534	46,849
F52	2023	21,056	4,957	1,199	3,127	32,817	0.2798	2,734	690,979	162,686	39.35	102,612
F53	2021	35,343	4,544	1.49	1,202	31,855	0.4106	2,798	1125,853	144.75	47,458	38,285
F53	2022	33,896	3,426	0.523	1,188	31,916	0.358	2.26	1081.82	109,346	16,688	37,914
F53	2023	24,748	1,757	1,406	1,519	31.99	0.2508	1,408	791,691	56,202	44.97	48,583
F54	2021	12,173	3.07	1,212	2.28	28,058	0.1714	2,302	341,563	86,138	34,003	63,974
F54	2022	33,396	1,658	1,213	1,141	28,093	0.3947	1,992	938,168	46,591	34,075	32,063
F54	2023	31,074	3,739	0.319	1,651	28.16	0.3587	2,198	875,049	105.28	8,971	46,489
F55	2021	12,357	3,088	0.552	2,205	28,166	0.1529	2,351	348,057	86,968	15,552	62,116
F55	2022	15,024	1,918	1,509	2,525	28,188	0.2002	2,066	423,509	54,072	42,542	71,171
F55	2023	36,129	4,356	1,823	1,681	28,304	0.4348	2,611	1022.59	123,301	51,606	47,571
F56	2021	29,307	3,606	1.69	1,088	30,262	0.3172	2,222	886,886	109,112	51,143	32,917

firm	year	esg_lag	vaic	der	cr	size	road	tobinsq	esg_size	vaic_size	der_size	cr_size
F56	2022	36,154	2,377	2,154	1,675	30,293	0.388	2,495	1095,181	72,001	65,254	50,736
F56	2023	24,946	3,721	1,666	1,425	30,421	0.2894	2,405	758,882	113,211	50,696	43,364
F57	2021	30,407	1,789	0.358	1,493	28,998	0.3353	2,204	881,729	51,871	10,388	43,284
F57	2022	32,281	3,202	0.957	1,753	29,077	0.33	2,902	938,639	93,104	27,835	50,983
F57	2023	31,758	3,447	1.96	2,972	29,088	0.3627	1,875	923,779	100,258	56,999	86,439
F58	2021	16.32	3,992	1.5	2,586	32,562	0.218	2,458	531,404	129,982	48,842	84.21
F58	2022	15.39	2.45	2,146	1,316	32,633	0.1362	2,052	502,239	79,957	70,019	42,953
F58	2023	36,122	3,281	1,278	1,726	32,673	0.3904	2,001	1180,229	107,207	41,761	56,401

